BSCS FINAL PROJECT

ReMIND: Generative AI Based Memory Rebuilder

Project Advisor

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Presented by
Group ID: F25CS104

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Faculty of Information Technology & Computer Science

University of Central Punjab

Complete System

SDP Phase IV

ReMIND: Generative AI based Memory Rebuilder

Advisor: Muhammad Zulkifl Hasan

Team F25CS0104

Member Name	Primary Responsibility
M. Hassan Ashraf	AI, Research and Data Analysis
M. Talha Faizan	Frontend and APIs
Mubashar Tanveer	Backend and Testing

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Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

F25 Idea Defence Feedback

PO Project Office

Reply Reply all Forward

To: L1F22BSCS0086- MUHAMMAD HASSAN ASHRAF; L1F22BSCS0103- MUHAMMAD TALHA FAIZAN; **+1 other**

Mon 01/09/2025 10:35

Cc: Dr Adnan Ghafoor; Muhammad Zulkifl Hasan

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	**CS-15**
Project Title	**GENERATIVE AI BASED MEMORY REBUILDER**

Similar products already exist. If the patient/a family member also does not have the details of event which we want to restore. Then system cannot perform.

Problem statement not clear. Scope not clear.

Students are incapable of implementing the idea. They don't know the basics of the project. Better to change the idea.

Major revision required. Revise your project idea in light of comments provided. Last date to submit the revised idea form is **Thursday, September 11, 2025, before 3:00 PM** in Project Office (F-304 Building A).

Phase 2 (SDS) Feedback (Screenshot)

PO **Project Office**

To: L1F22BSCS0086- MUHAMMAD HASSAN ASHRAF;
L1F22BSCS0103- MUHAMMAD TALHA FAIZAN;
L1F22BSCS0094- MUBASHAR TANVEER

Thu 30/04/2026 13:38

Cc: Muhammad Zulkifl Hasan

Dear Students,

Following is the feedback regarding your phase 2 (SDS):

F25CS104	**REMIND: GENERATIVE AI BASED MEMORY REBUILDER**

Redraw class diagram. Should be online deployed. Must compare with at-least one available app.

Workflow knowledge was missing.

Abstract

ReMIND is an AI assisted web application designed to support individuals experiencing memory loss disorders such as Alzheimer's disease, dementia, and age-related cognitive decline. Memory loss affects not only a person's ability to recall past events but also their emotional stability, relationships, and overall sense of identity. Most existing digital solutions rely heavily on manual input, and when no prior record or recollection exists, they fail to assist patients and caregivers in restoring meaningful experiences. ReMIND addresses this limitation by offering semi-automated memory reconstruction that helps users reconnect with their personal history.

The system collects and analyzes different forms of data, including written notes, photographs, and voice recordings, and organizes them into personalized timelines that reflect moments from daily life. Through artificial intelligence, ReMIND generates contextual details that can suggest likely places, people, and events related to the user's memories. This enriched information helps patients retrieve forgotten experiences in a more natural and emotionally resonant way. The platform creates an interactive environment where patients and caregivers can revisit, discuss, and refine reconstructed memories together, strengthening emotional connection and supporting cognitive engagement.

The project brings together concepts from artificial intelligence, natural language processing, computer vision, speech processing, data management, and human centered design. The focus is not solely on automation but on creating an experience that prioritizes empathy, accessibility, and user comfort. By transforming fragmented or incomplete data into meaningful memory narratives, ReMIND aims to restore a sense of identity and continuity for individuals facing cognitive decline.

The expected outcome is a functional prototype that demonstrates how intelligently guided memory reconstruction can contribute to cognitive rehabilitation and long-term digital memory preservation. ReMIND highlights the potential of emerging AI systems to support emotional wellbeing and provide hope for individuals whose memories are fading.

1. Introduction

ReMIND is an AI-assisted web application designed to help individuals experiencing memory loss, including patients affected by Alzheimer's disease, dementia, and age-related cognitive decline. The system reconstructs fragmented memories using multimodal inputs such as text, images, and voice recordings. By combining artificial intelligence with an elderly-friendly user experience, ReMIND generates meaningful memory narratives that help patients re-live past experiences and strengthen emotional connections with caregivers and family members. The project is motivated by the emotional, cognitive, and social challenges associated with memory loss and aims to provide accessible technological support through intelligent memory reconstruction.

1.1 Product

ReMIND is an intelligent web-based memory reconstruction platform that assists individuals suffering from memory-related disorders by transforming fragmented information into meaningful memories. The system accepts multimodal inputs including text descriptions, photographs, and voice recordings, which are processed through integrated artificial intelligence services. Using speech recognition, image enhancement, contextual understanding, and narrative generation techniques, ReMIND reconstructs memory artifacts that users can revisit through an interactive chat-based interface. The platform also supports reminders, memory timelines, caregiver collaboration, secure authentication, and cloud-based data management. Through its combination of AI-powered reconstruction and human-centered design, ReMIND helps bridge emotional and cognitive gaps while improving memory recall and quality of life.

1.2 Background

The project lies at the intersection of artificial intelligence, cognitive rehabilitation, human-computer interaction, and healthcare technology. It applies modern AI techniques including natural language processing, speech processing, image enhancement, and generative AI to support individuals affected by memory decline. By integrating multiple AI services into a unified platform, ReMIND aims to improve memory recall, emotional well-being, and caregiver involvement while demonstrating the potential of intelligent systems in cognitive support and digital healthcare solutions.

1.3 Objective(s)/Aim(s)/Target(s)

The primary objective of ReMIND is to design, develop, and deploy an AI-assisted memory reconstruction platform that enables individuals experiencing memory loss to reconstruct meaningful memories from fragmented multimedia information. The project combines artificial intelligence, accessibility, and empathetic design principles to provide cognitive support for patients while enabling caregivers to participate in the memory reconstruction process.

The specific objectives are as follows:

Memory Reconstruction: Transform incomplete data such as text notes, photos, and voice recordings into coherent and contextually rich memory narratives.

Elderly-Friendly Interface: Design a simple, accessible interface optimized for older users and non-technical caregivers.

Secure Data Handling: Implement encryption, authentication, and cloud storage to protect sensitive personal data.

Proof of Concept Demonstration: Deliver a functional prototype using existing AI APIs and web technologies to validate the concept.

Overall, ReMIND seeks to enhance emotional well-being, cognitive support, and human connection through accessible AI innovation.

1.4 Scope

The scope of ReMIND includes the design, implementation, testing, and deployment of a web-based memory reconstruction system that processes text, image, and audio inputs to generate meaningful memory narratives. The system incorporates AI-powered memory reconstruction, multimedia enhancement, contextual analysis, reminder management, caregiver collaboration, user authentication, cloud storage integration, and timeline generation. The platform is designed with accessibility considerations for elderly users and supports secure handling of personal data. ReMIND is developed as a proof-of-concept solution intended to demonstrate the practical feasibility of AI-assisted memory reconstruction. Medical diagnosis, hospital information system integration, and real-time biometric monitoring remain outside the project scope.

1.5 Business Goals

As a successfully implemented proof-of-concept platform, ReMIND demonstrates the practical feasibility of AI-driven memory reconstruction for individuals experiencing memory loss. The project validates how modern AI technologies can be integrated to provide emotional and cognitive support through memory rebuilding, contextual recall, and caregiver collaboration. ReMIND establishes a foundation for future research, commercialization, and clinical adaptation within healthcare and assisted-living environments. By combining accessibility, empathy, and artificial intelligence, the project highlights opportunities for innovative digital health solutions that improve patient well-being and strengthen human connections through preserved memories.

1.6 Document Conventions

The Document Conventions being used throughout the document are following:

- Font Style being used is Times New Roman.
- Main Headings are of Size 16.
- Sub-Headings are of Size 14.
- Content/Paragraph and Tertiary headings are of Size 12.
- Auto line spacing is used.

2. Technical Architecture

ReMIND is a specially designed web application that makes use of both **open-source AI models** and **Commercial Off-The-Shelf (COTS) AI services**. The system incorporates specialized technologies for natural language processing, speech recognition, voice synthesis, image enhancement, and contextual understanding rather than creating machine learning models from the ground up. This strategy allowed for quick development while guaranteeing dependable performance and excellent results within the technical and academic limitations of the project. ReMIND supports online, nearly real-time processing through a **layered client-server architecture**. User authentication, multimedia uploads, memory reconstruction requests, timeline creation, reminder management, and caregiver interactions are among the event-driven transactions that the system manages. The system keeps track of audit logs, feedback records, and evaluation metrics for administrative and monitoring purposes in addition to transactional processing.

Key Application Components and Duties:
The **Frontend Layer uses React** to create a browser-based user interface. Users can access timelines, manage reminders, upload multimedia content, engage with reconstructed memories, and communicate via a chat-based interface. With usability considerations for senior users and caregivers, the interface is responsive and **optimized for desktop and mobile devices**.

FastAPI is used to implement the Backend Layer, which acts as the platform's central control unit. The AI Processing Layer consists of specialized Text, Voice, and Image Processing modules. Memory reconstruction, retrieval-augmented generation, contextual understanding, and natural language processing are all performed by the Text Processing Module. Speech-to-text conversion, emotion and tone analysis, and voice synthesis are all performed by the Voice Processing Module. Image enhancement, face restoration, image caption generation, and visual context extraction are all performed by the Image Processing Module.

Both structured and unstructured data produced throughout the system are managed by the Data Layer. The **main database** used to store user accounts, memory artifacts, timelines, audit logs, feedback, reminders, and metadata is **PostgreSQL**.

The finished architecture effectively combines database management, cloud storage, AI modules, frontend services, and backend processing into a single ecosystem that can reconstruct meaningful memories from multimodal inputs. Beyond the proof-of-concept phase, the modular design facilitates future expansion, scalability, and maintainability.

2.1 Application and Data Architecture

Component Diagram:

This Component Diagram illustrates the physical architecture of the ReMIND system, showing how thesoftware components interact. The core communication hub is the API Gateway, which routes requestsbetween different modules. The Frontend (React.js) provides user interfaces like the Memory Input UIand Reconstructed Memories Viewer. The Backend (FastAPI) handles User Management andEvent Clustering. The system relies on external AI Services for tasks like Story Generation and Speech-to-Text. Data is persistently stored in a PostgreSQL Database and Cloudinary Media Storage.The Notification Module handles reminders, and the entire system is deployed using VPS Hosting via Hostinger.

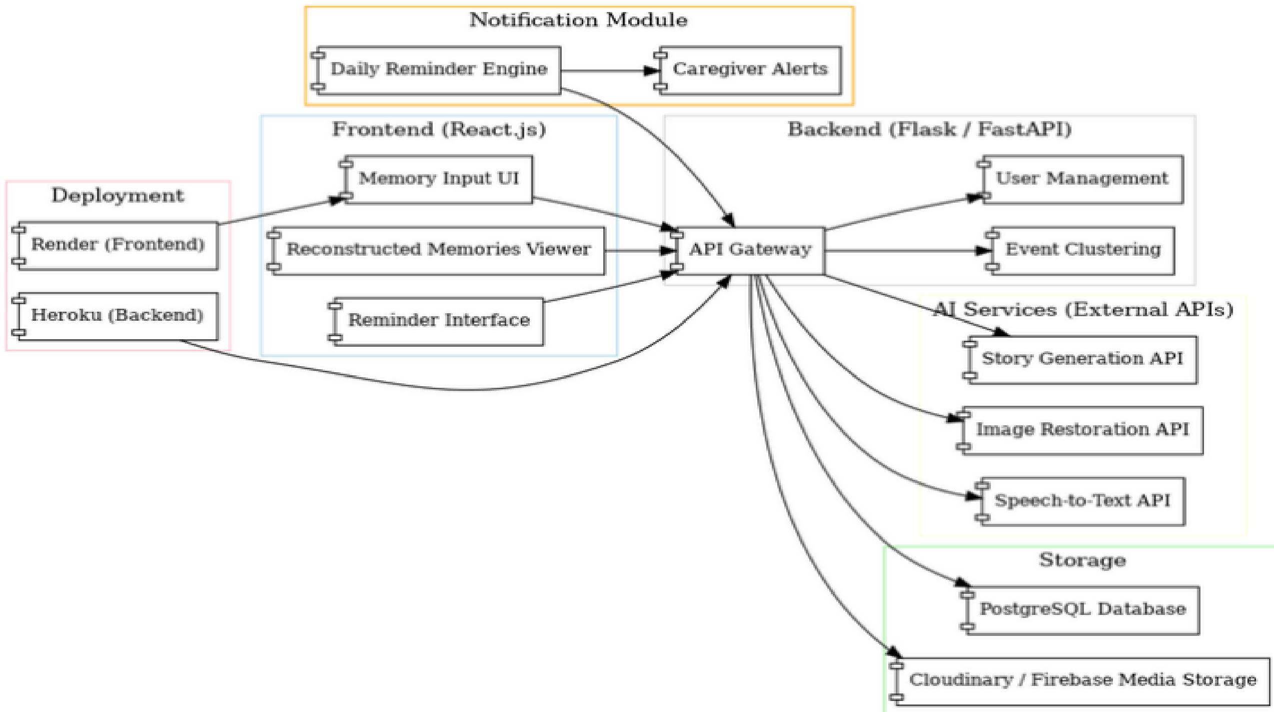


Fig 2.1.1: Component Diagram

ERD Diagram:

This Entity-Relationship Diagram (ERD) visualizes the structure of a database for ReMIND. It models key entities such as the adaptable User (specialized as Patient, Caregiver, and Admin) and the relationships that connect them. Patients, associated with Conditions, receive care and are managed by Caregivers. The operational core includes Chat Sessions for communication, composed of Messages and Media Files, with support for Reminders (which issue Notifications), and the capture of user Feedback. All critical activities are logged in an Audit Log, and the system includes functionality for processing media using AI Processing Tasks.

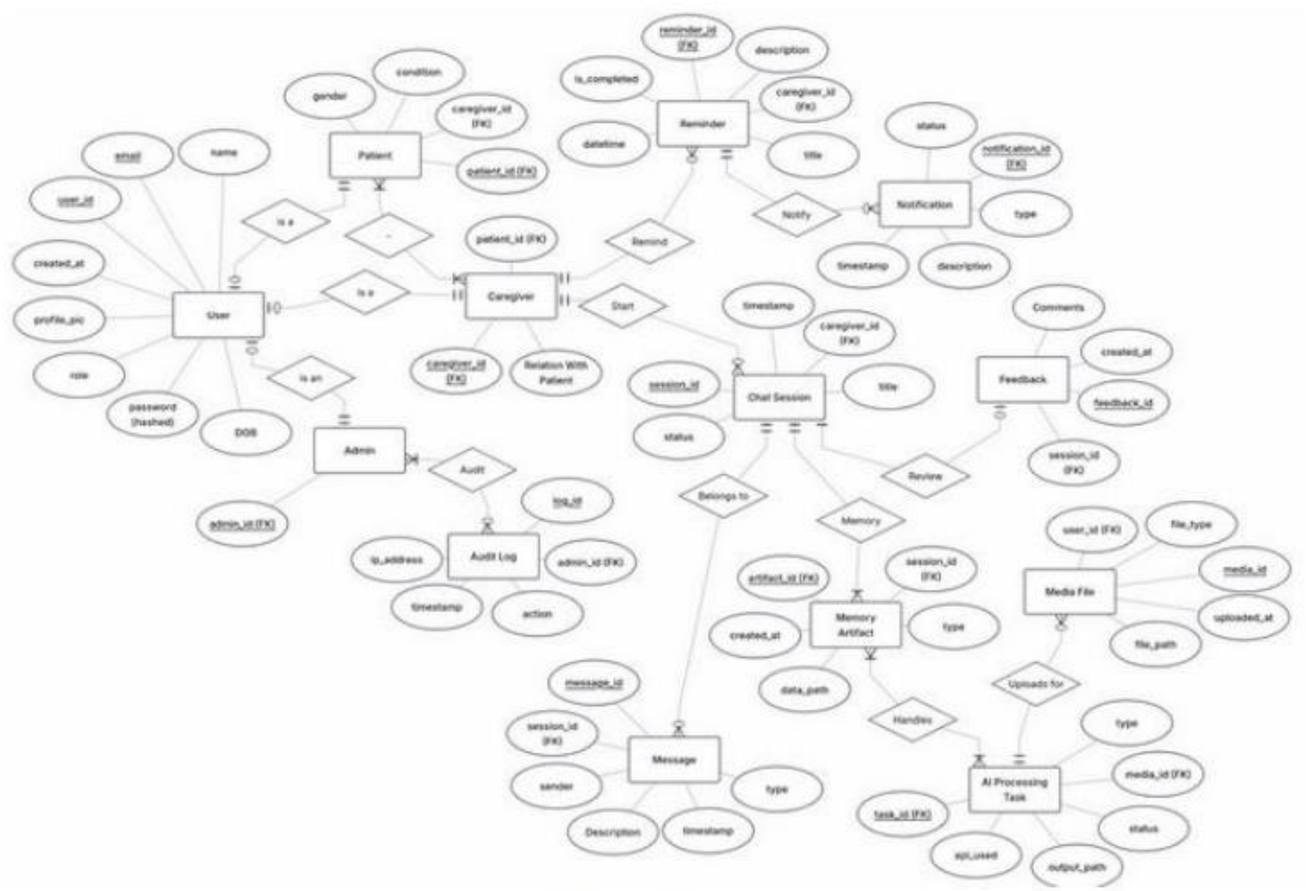


Fig 2.1.2: ERD Diagram

UML Class Diagram:

This UML Class Diagram models the structure of ReMIND. It organizes functionality into modules: Profiles, Audit Log, Caregiver & Patient, Chat, and AI & Media. The Profiles module defines User, specialized in Caregiver Profile, Patient Profile, and Admin Profile. The Caregiver & Patient Module manages Reminders and Notifications. The Chat Module handles Chat Sessions, Messages, and Feedback. Finally, the AI & Media Module defines Media, Memory Artifact, and AI Processing Task for handling media content.

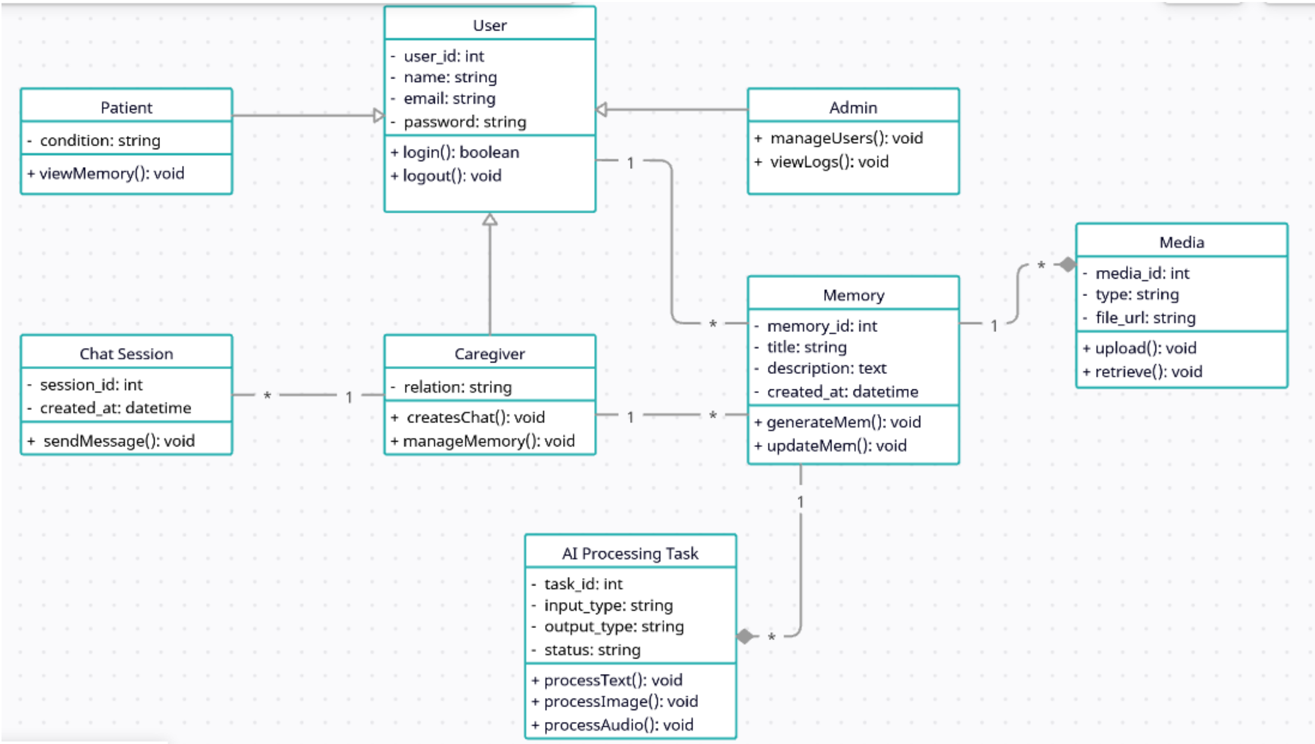


Fig 2.1.3: UML Class Diagram

Relational Schema:

This relational schema, based on the ReMIND-DATABASE ERD. It features a central User table, with specialized child tables: AdminProfile, CaregiverProfile, and PatientProfile, linked by foreign keys (FK) on the respective ID fields. The core interactions occur in the ChatSession table, which references CaregiverProfile and PatientProfile. Communication data is stored in Message and Media tables. Support functionality includes Reminder and Notification tables. All actions are tracked in the Auditlog table, while AIProcessingTask and MemoryArtifact handle media processing and retrieval.

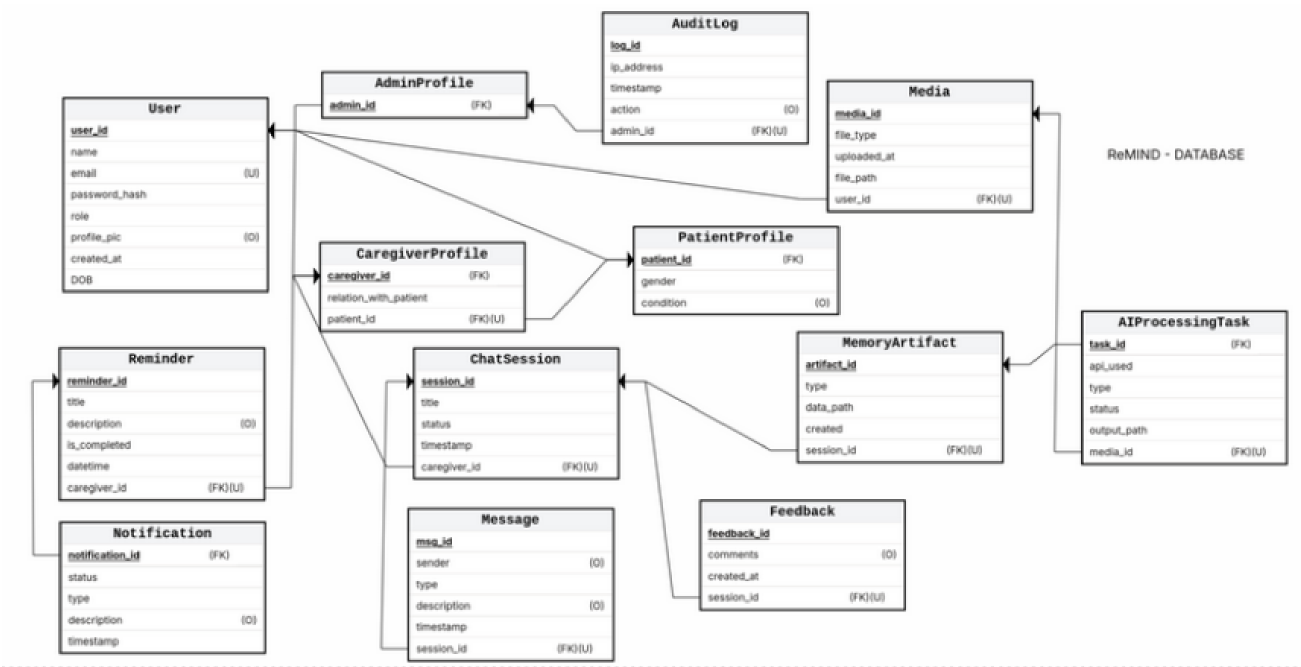


Fig 2.1.4: Relational Schema Diagram

Decision Table:

Rule	Input	Alternative Conditions	Action	Output
R1	User provides valid login credentials	Invalid credentials	Authenticate user, create session	Access granted / error message displayed
R2	Patient uploads multimedia input	Unsupported format or missing metadata	Accept input, store temporarily, trigger preprocessing	Acknowledgment / error message
R3	Uploaded file validated successfully	File corrupted or exceeds size limit	Proceed to AI preprocessing and enhancement	Process initiated / warning prompt
R4	Image input detected	No image detected	Trigger Real-ESRGAN enhancement and tagging	Enhanced image + metadata stored
R5	Audio input detected	No audio provided	Send to Whisper API for transcription	Transcript text returned
R6	Text or transcript available	Missing or empty text input	Send to GPT-4 for contextual understanding	Structured text/context response
R7	All media processed successfully	Any AI module fails	Aggregate data and reconstruct event	Reconstructed memory / retry message
R8	Narrative generation successful	AI fails to generate or timeout	Store reconstructed memory in repository	Text + voice response / system retry
R9	Reminder set by caregiver	Reminder missing or invalid	Save reminder in database	Reminder confirmation / validation error

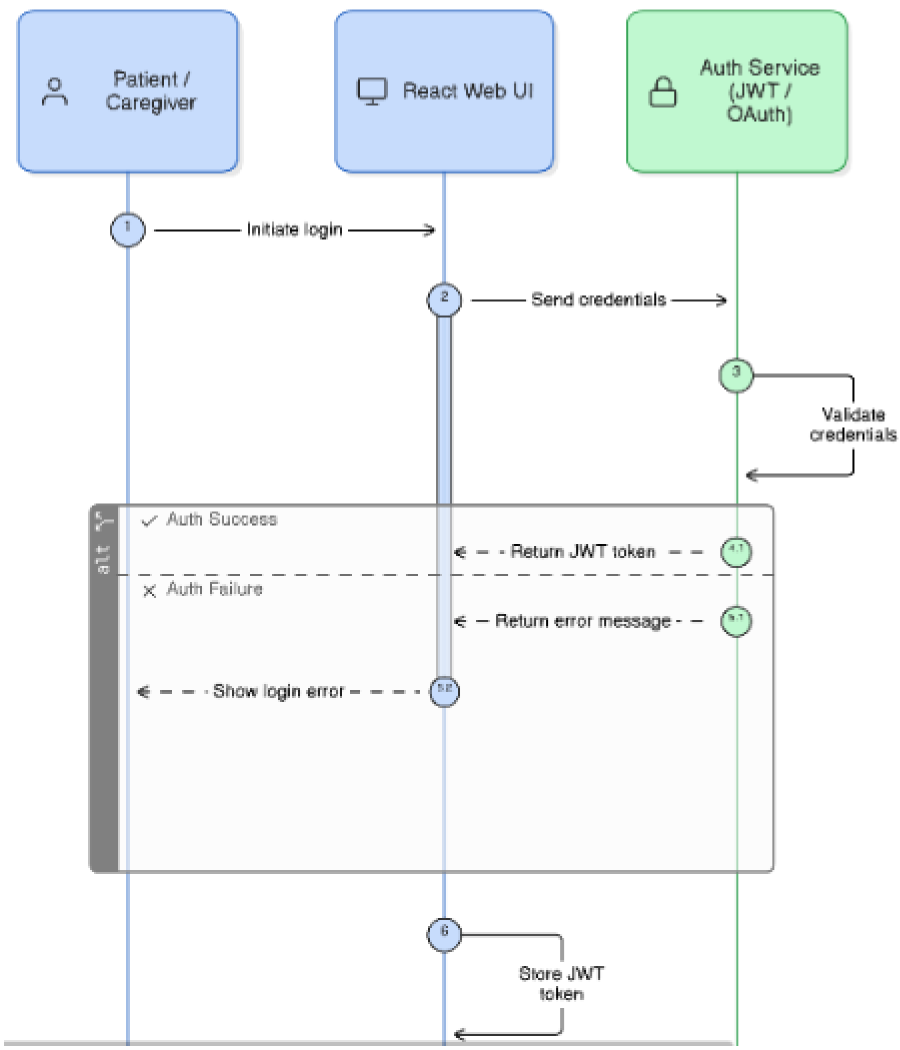
R10	Feedback submitted by patient/caregiver	Feedback skipped	Store feedback in log	Confirmation message / skip noted
R11	Admin monitors system logs	Logs not accessible	Display system status dashboard	Log summary / permission error

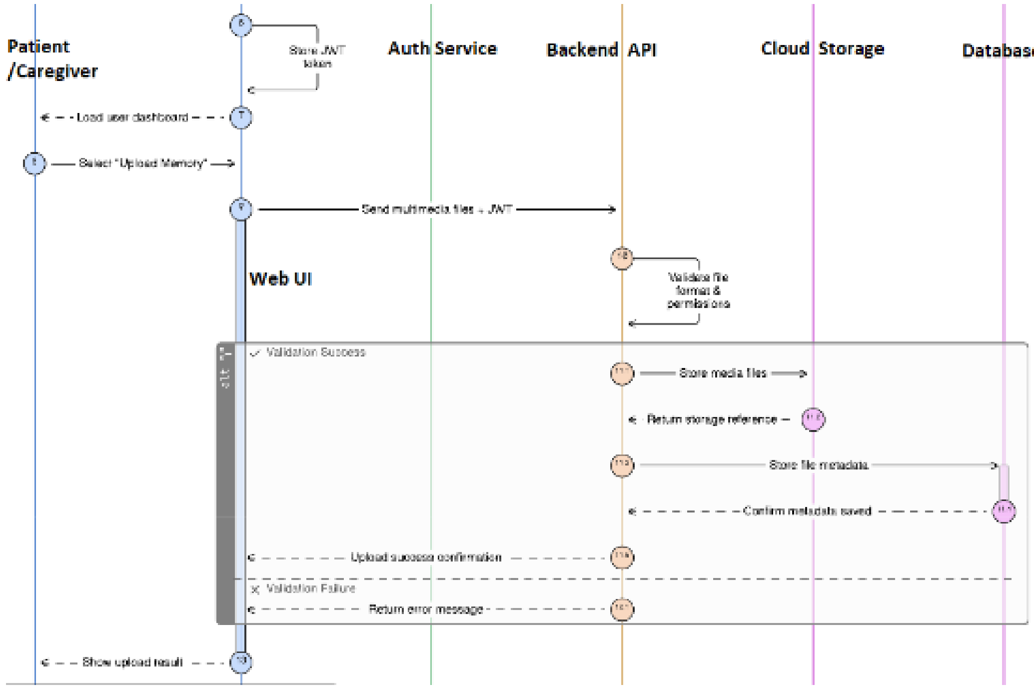
Table 2.1.5: Decision Table

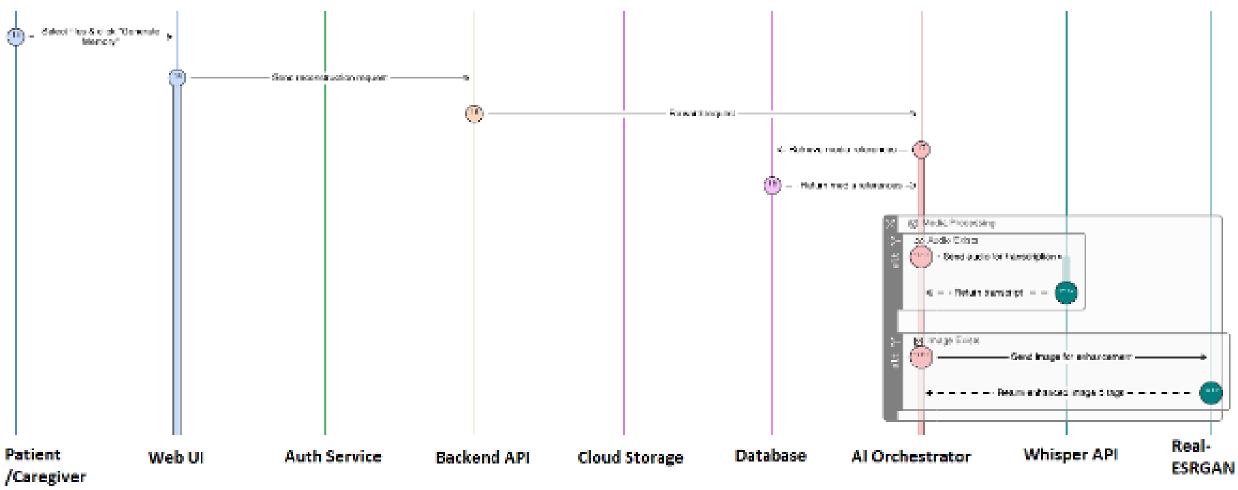
2.2 Component Interactions and Collaborations

Design Sequence Diagram:

The process flow begins with User Authentication, where the Frontend UI communicates with the Auth Service to validate credentials. Following login, the diagram maps the lifecycle of a memory:
The Frontend sends data to the Backend (Flask/FastAPI), which coordinates with Cloud Storage (Cloudinary) for media and triggers AI Services (like gTTS for speech-to-text or Groq for analysis). These services process the data and return it to the Memory Reconstruction Engine. Finally, the rebuilt memory the Database and delivered back to the Frontend for the user to view.




Patient /Caregiver
Auth Service
Backend API
Cloud Storage
Database
Store JWT token
Load user dashboard
Select "Upload Memory"
Send multimedia files + JWT
Web UI
Validate file format & permissions
Validation Success
Store media files
Return storage reference
Store file metadata
Confirm metadata saved
Upload success confirmation
Validation Failure
Return error message
Show upload result


Patient /Caregiver
Web UI
Auth Service
Backend API
Cloud Storage
Database
AI Orchestrator
Whisper API
Real-ESRGAN

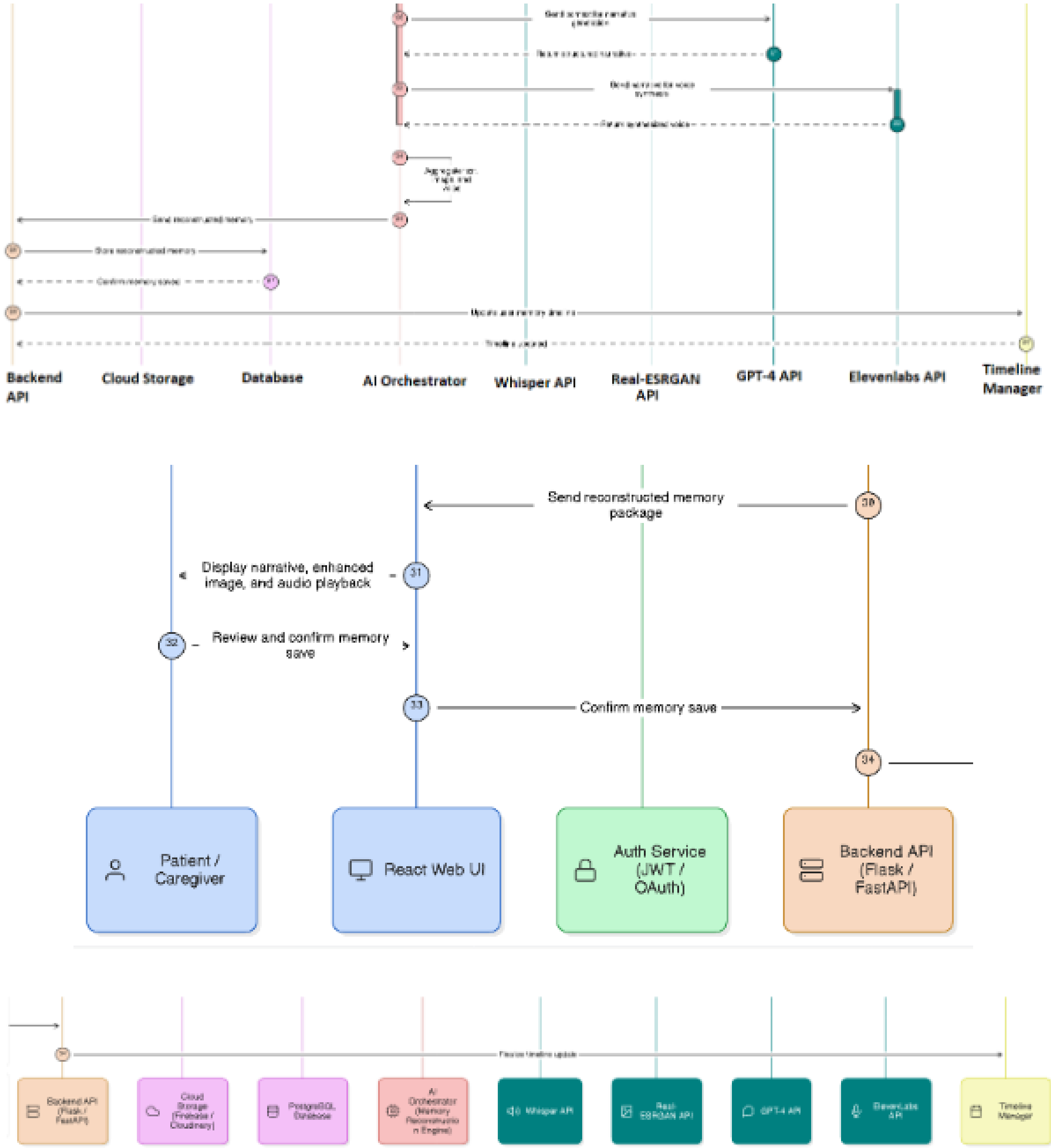


Fig 2.2.1: Sequence Diagram

Collaboration Diagram:

This Collaboration Diagram outlines the functional flow of the ReMIND system. It begins with the User initiating uploadMemory() (1) through the Frontend, which triggers sendJSON() (2) to the API Gateway. After the Auth Service performs authenticatesUser() (3), the MPC manages the Memory Reconstruction phase by identifying the input type (5) for the Te xt, Image, or Audio AI Engines. Once processed, the reconstructedOutput() (6) is sent to the Evaluation Engine to calculateAccuracy () (7) before being committed to the Database. The cycle completes as the system storeLogs() (8) and transmits a requestJSON() (9) back to the Frontend for final displayMemory() (10) via the Display Output module.

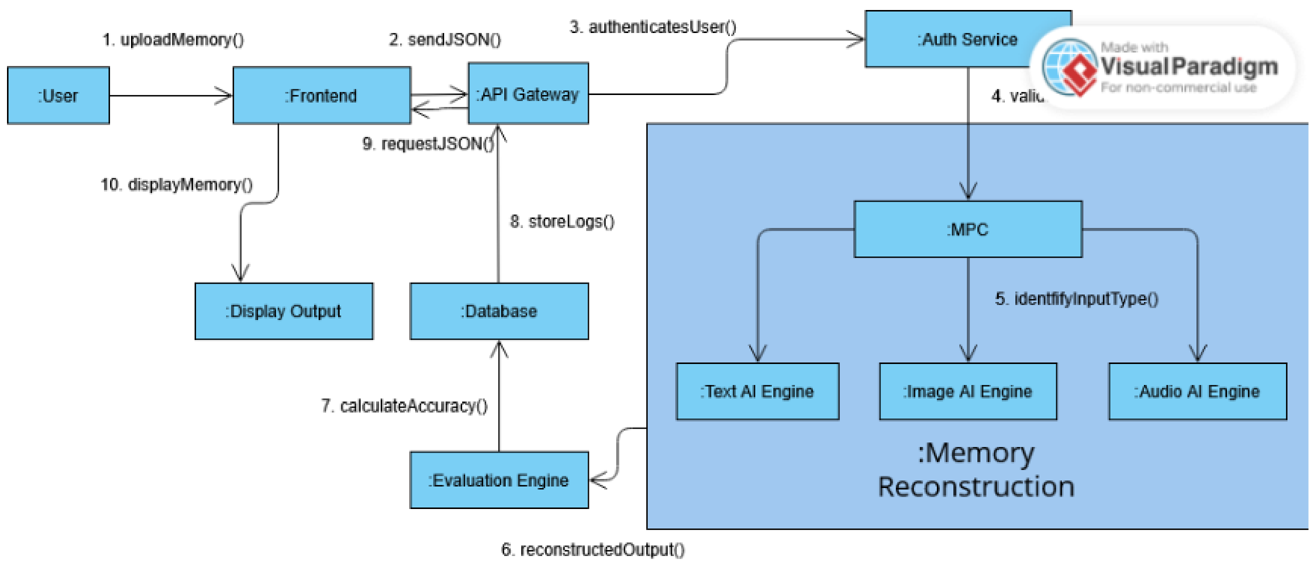


Fig 2.2.2: Collaboration Diagram

Activity Diagram:

The Activity Diagram illustrates the workflow of the ReMIND system for memory management and reconstruction. The process begins when the user logs into the system. Upon successful authentication, the user is required to complete a set of security questions, including both multiple-choice and text-based responses, to verify identity. After verification, the user uploads or enters memory-related information, which may include text, images, audio, or other multimedia content. The system then performs memory analysis and processing to extract meaningful information from the provided data. Once processed, the memory is stored and tagged with relevant categories and metadata to facilitate efficient organization and retrieval. Finally, the user can access and view the reconstructed or stored memories through the memory retrieval and display module, after which the workflow concludes.

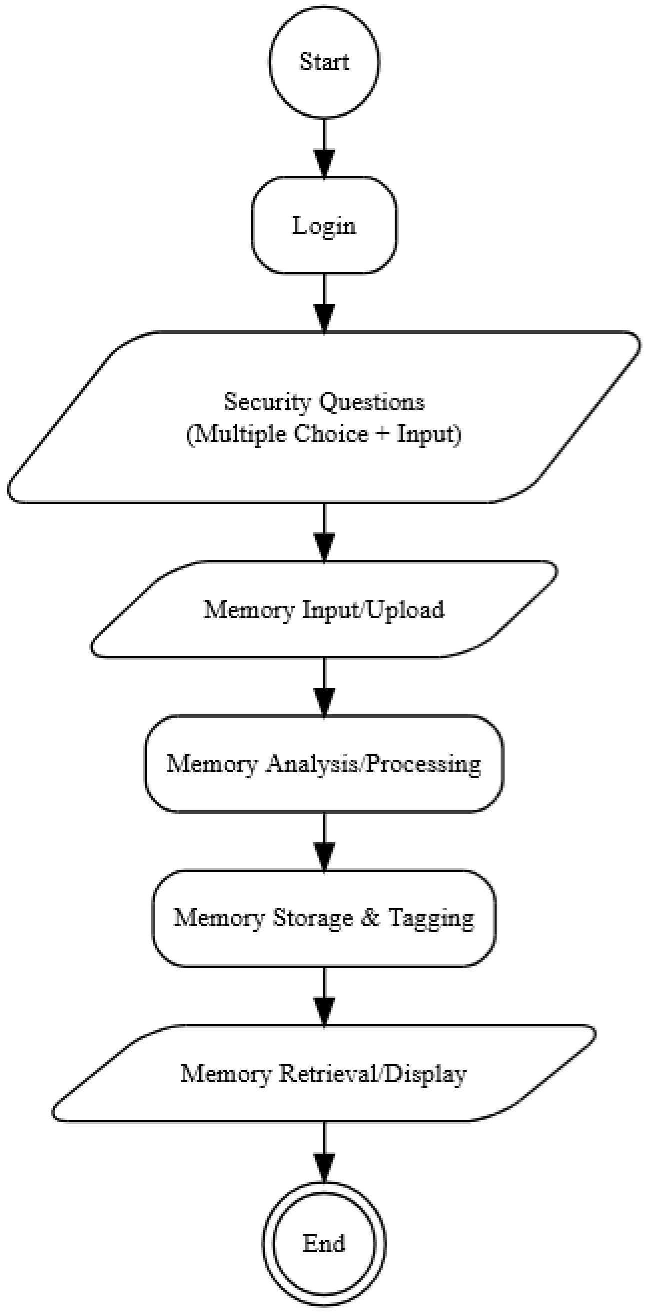


Fig 2.2.3: Activity Diagram

Event Table:

Event	Event Name	Trigger Type	Source	System Activity	Output
E1	User Login	External (User Action)	Patient / Caregiver / Admin	Authenticate credentials, create session	Dashboard or role-based access
E2	Media Upload	External (User Action)	Patient / Caregiver	Validate and store uploaded media	Temporary file storage + confirmation
E3	Data Preprocessing	Internal (System)	Backend	Cleans, formats, validates input data	Ready media for AI modules
E4	Image Enhancement	External (AI Service)	Real-ESRGAN	Enhances image and extracts context	Enhanced image file stored
E5	Audio Transcription	External (AI Service)	Whisper API	Converts audio to text, detects tone	Transcript returned
E6	Text Analysis	External (AI Service)	GPT-4	Interprets and structures text context	Semantic data created
E7	Event Reconstruction	Internal (AI Process)	AI Module	Combines multimodal data into cohesive memory	Structured narrative generated
E8	Voice Synthesis	External (AI Service)	ElevenLabs	Converts narrative text to natural audio	Voice file generated and stored
E9	Memory Storage	Internal (Database)	Backend → PostgreSQL	Saves reconstructed memory with metadata	Record created in memory repository
E10	Memory Retrieval	External (User Request)	Patient / Caregiver	Fetches requested memory or timeline	Displayed in frontend interface
E11	Reminder Creation	External (User Action)	Caregiver	Adds reminder entry in database	Confirmation sent to user
E12	Reminder Notification	Internal (Timed Event)	Notification Module	Sends scheduled alert to patient	Notification on UI
E13	Feedback Submission	External (User Action)	Patient / Caregiver	Logs ratings and suggestions	Stored in feedback log
E14	System Monitoring	Internal (Admin Action)	Administrator	Reviews logs, checks API integrations	Diagnostic report generated

Table 2.2.4: Event Table

Detailed DFD with Explanation:

DFD Level 1:

The DFD Level 1 diagram illustrates the primary functional processes of the ReMIND system and the flow of data between users, databases, and AI services. The process begins with User Authentication (3.0), where Patients, Caregivers, and Administrators provide login or registration information. The system validates these credentials through the Accounts Database and grants authorized access to subsequent modules. Once authenticated, users can interact with the Data Processing module (1.0), which receives and validates uploaded multimedia and textual information. The processed data is stored in the Data Processing Database and forwarded to the Memory Reconstruction module (2.0). This module utilizes AI-powered services to analyze the processed information, reconstruct meaningful memories, and generate personalized results and feedback. The reconstructed memories are then stored in the Memory Artifacts Database for future retrieval, while analytical reports and system statistics are made available to administrators for monitoring and management purposes.

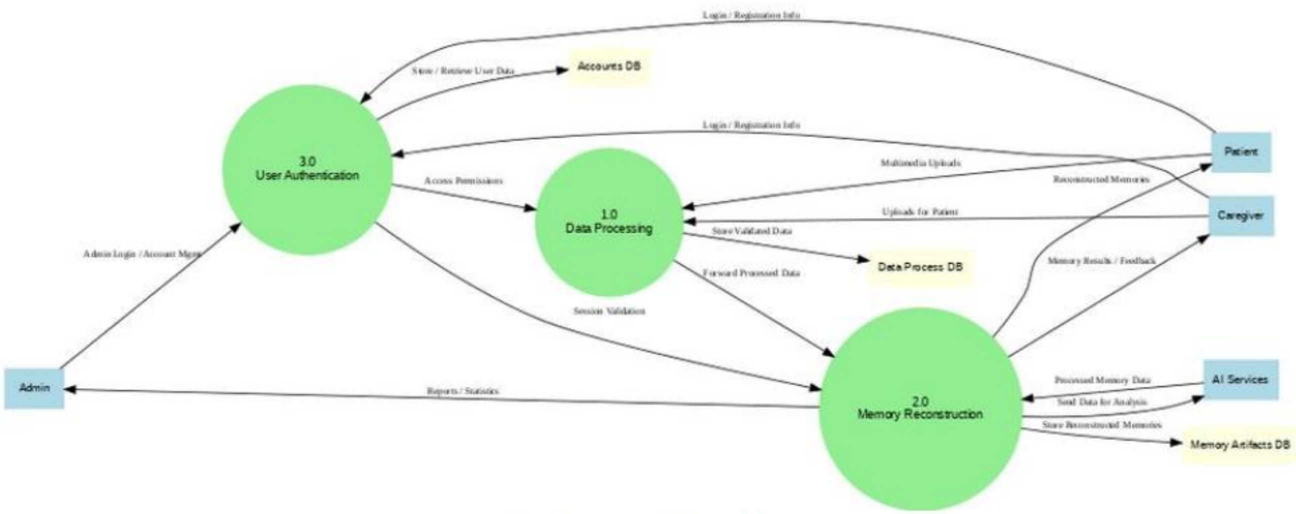


Fig 2.2.5: DFD Level 1

DFD Level 2:

The DFD Level 2 diagram provides a more detailed view of the internal operations of the ReMIND system by decomposing the Memory Reconstruction process into smaller functional components. The system begins with User Authentication (3.0), which validates user credentials and manages access permissions for Patients, Caregivers, and Administrators. Data Processing (1.0) receives multimedia and textual inputs from users, performs validation and preprocessing, and stores the processed information in the Data Processing Database. The processed data is then forwarded to the Memory Reconstruction module and its sub-processes for further analysis. Within Memory Reconstruction, the API Calls & Integration process (2.0.1) manages communication with external AI services. It sends relevant media and data for analysis, receives AI-generated outputs, and records all API interactions in the APIs Database for tracking and auditing purposes. The generated results are then utilized by the Memory

Verification process (2.0.2), which evaluates and validates reconstructed memories using user-provided multimedia content. Verified memories are stored in the Memory Artifacts Database, while verification outcomes, feedback, and monitoring information are provided to the Administrator. This decomposition offers a clearer representation of how AI integration and verification mechanisms contribute to accurate memory reconstruction within the ReMIND system.

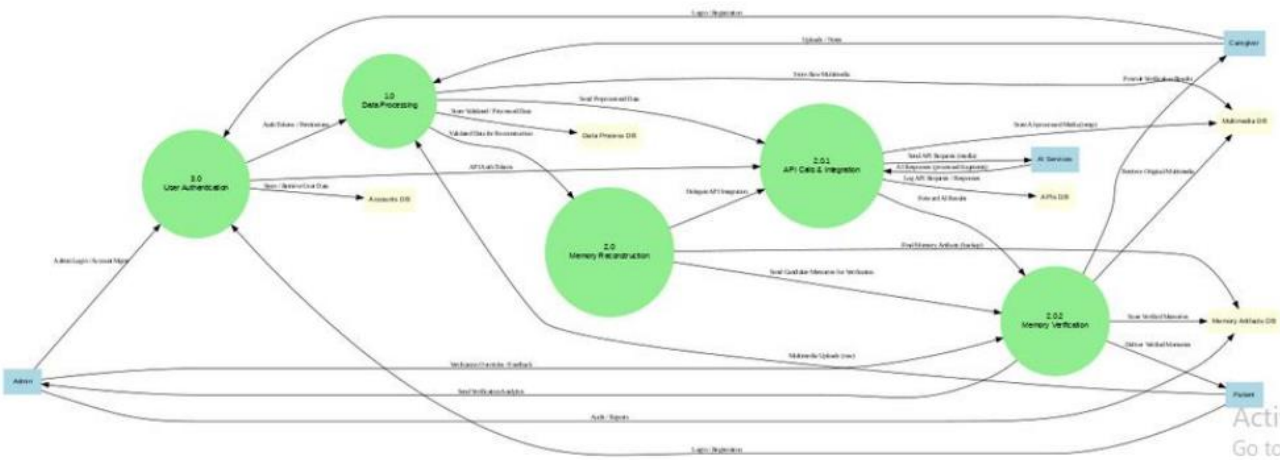


Fig 2.2.6: DFD Level 2

2.3 Design Reuse and Design Patterns

The development of ReMIND extensively utilizes software reuse principles to improve reliability, accelerate development, and ensure technical feasibility within the project's academic scope. Rather than developing every component from scratch, the system integrates well-established frameworks, libraries, cloud services, and pre-trained artificial intelligence models. This approach allowed the development team to focus on system integration, workflow orchestration, user experience, and memory reconstruction logic while leveraging proven technologies for supporting functionalities.

At the application layer, ReMIND reuses modern open-source frameworks including React.js for frontend development and FastAPI for backend services. These frameworks provide standardized routing, state management, API handling, middleware support, and security mechanisms that significantly reduce development complexity while improving maintainability and scalability.

At the intelligence layer, the system integrates pre-trained artificial intelligence models and external AI services rather than developing custom machine learning models. The Text Processing Module utilizes Groq-Llama 3 and Retrieval-Augmented Generation (RAG) techniques for memory reconstruction and contextual understanding. The Voice Processing Module leverages ElevenLabs STT for speech processing and Google TTS voice synthesis, while the Image Processing Module utilizes GFPGAN, Real-ESRGAN, and Blip-lightweight for image enhancement, face restoration, and image caption generation. Reusing these mature AI technologies eliminated the need for extensive model training and optimization while ensuring high-quality outputs.

At the data layer, PostgreSQL is reused as a robust relational database management system for storing user information, memory artifacts, reminders, feedback, audit logs, and metadata. Cloudinary is utilized for secure cloud-based storage and delivery of multimedia assets. SQLAlchemy and Alembic are reused for database interaction and schema migration management.

Several software engineering and architectural design patterns were adopted throughout the implementation. The Layered Architecture Pattern separates the Presentation, Application, AI Processing, and Data layers to improve modularity and maintainability. The API Gateway and Service-Oriented approach is used within the backend to coordinate communication between frontend services, AI modules, databases, and cloud storage platforms. The Model-View-Controller (MVC) concept influences the separation of user interface, business logic, and data management responsibilities. Additionally, middleware-based authentication and authorization mechanisms are implemented to provide secure access control throughout the system.

Development and collaboration tools such as GitHub, Postman, Figma, and modern package management systems were also reused as part of standard software engineering practices. Overall, ReMIND combines reused technologies with original system design, workflow integration, data modeling, and memory reconstruction logic, resulting in a functional and scalable proof-of-concept platform.

2.4 Technology Architecture

Data and Processing Characteristics

ReMIND collects, processes, and manages both structured and unstructured data generated during memory reconstruction workflows. The system maintains user credentials, authentication information, role assignments, memory artifacts, reminders, timelines, feedback records, audit logs, and AI-generated outputs. Additionally, the platform manages multimedia inputs including text notes, images, and voice recordings submitted by users for memory reconstruction.

The system operates in an online, event-driven environment where user interactions trigger near real-time processing workflows. When multimedia content is uploaded, specialized AI modules perform contextual analysis, speech processing, image enhancement, and memory reconstruction before generating user-facing outputs. Processing is primarily transactional in nature, while system logs, evaluation of metrics, and feedback records provide basic analytical reporting and monitoring capabilities.

Technology Stack and Platforms:

- **Programming Languages:** Python, JavaScript
- **Frontend Technologies:** React.js, React Router, Context API, Axios, CSS3
- **Backend Technologies:** FastAPI, RESTful APIs, SQLAlchemy, JWT Authentication, OAuth2
- **Text Processing Technologies:** Groq Llama 3, Retrieval-Augmented Generation (RAG), FAISS, Sentence Transformers
- **Voice Processing Technologies:** ElevenLabs STT, PyDub, Librosa, Google Text-to-Speech
- **Image Processing Technologies:** GFPGAN, Real-ESRGAN, Blip
- **Database Platform:** PostgreSQL, Alembic
- **Media Storage Platform:** Cloudinary
- **Version Control and Collaboration:** GitHub
- **API Testing and Validation:** Postman
- **UI/UX Design and Prototyping:** Figma
- **Deployment and Hosting:** Hostinger VPS deployment and hosting.
- **Development Hardware:** Personal laptops and cloud-based computing resources
- **End-User Interface:** Browser-based interface accessible through desktop and mobile devices

Interface and Network Architecture:

ReMIND provides a browser-based user interface accessible through modern web browsers on desktop and mobile devices. No specialized client software is required for system access. The platform follows a WAN-based architecture where users, caregivers, and administrators interact with the system remotely over the Internet.

Communication between the frontend, backend, database, cloud storage, and AI processing services occurs through secure RESTful APIs. This architecture enables seamless integration between system components while maintaining security, reliability, and accessibility for end users regardless of their physical location.

2.5 Architecture Evaluation

Frontend Technology Selection:

Selected Technology: React.js with CSS3

Reason for Selection:

React.js was selected because it provides a component-based architecture that simplifies the development of interactive and responsive user interfaces. Since ReMIND requires real-time user interaction, multimedia uploads, chat-based communication, role-based dashboards, and dynamic content rendering, React offers an efficient and scalable solution. CSS3 was used to create a responsive, modern, and elderly-friendly interface while maintaining full control over styling and accessibility features.

Advantages:

- Component-based architecture promotes code reusability and maintainability.
- Fast and responsive user experience through Virtual DOM rendering.
- Easy integration with RESTful APIs and backend services.
- Supports dynamic content updates without full page reloads.
- Large developer community and extensive ecosystem support.
- Responsive design support for both desktop and mobile devices.

Disadvantages:

- Requires understanding of modern JavaScript concepts and React lifecycle management.
- State management can become complex in large-scale applications.
- Initial project structure and setup require additional configuration compared to traditional HTML-based approaches.

Alternative Considered: Angular

Backend Technology Selection:

Selected Technology: FastAPI

Reason for Selection:
FastAPI was selected because it provides a lightweight, high-performance framework for building RESTful APIs while maintaining excellent compatibility with artificial intelligence and machine learning libraries written in Python. Since ReMIND integrates multiple AI modules, authentication services, database operations, and cloud-based resources, FastAPI enabled rapid development, clean architecture, and efficient communication between system components

Advantages:

- High-performance asynchronous request handling.
- Seamless integration with AI and data processing libraries.
- Automatic API documentation generation.
- Lightweight and modular architecture.
- Strong support for RESTful API development.
- Easy integration with authentication, database, and cloud services

Disadvantages:

- Fewer built-in features compared to larger frameworks.
- Requires manual configuration of some components.
- Smaller ecosystem compared to older Python frameworks.

Alternative Considered: Flask

AI Models and API Selection:
The ReMIND system utilizes specialized AI models and APIs for text processing, voice processing, and image enhancement. Instead of training custom machine learning models, the project integrates existing AI technologies to achieve high-quality results while remaining technically and computationally feasible

Groq Llama 3 (Text Processing and Memory Reconstruction):

Reason for Selection:
Groq Llama 3 was selected because it provides fast and context-aware text generation capabilities suitable for memory reconstruction. It enables the system to understand fragmented user inputs, generate coherent narratives, and support Retrieval-Augmented Generation (RAG) workflows while maintaining low response latency.

Advantages:

- Fast inference and response generation.
- Strong contextual understanding and reasoning capabilities are important.
- Suitable for conversational memory reconstruction.
- Easy integration with RAG pipelines.
- Open-Source Availability.

Disadvantages:

- Depends on external API availability.
- Generated responses may occasionally require validation.
- Performance is influenced by prompt quality and retrieved context.

Alternative Considered: OpenAI GPT-4

ElevenLabs Speech-to-Text (STT):

Reason for Selection:
ElevenLabs STT was selected because it provides accurate speech recognition and transcription capabilities across different accents and speaking styles. This is particularly important for elderly users who may have varied speech patterns and audio quality.

Advantages:

- High transcription accuracy.
- Handles different accents effectively.
- Supports natural conversational speech.
- Easy API integration.
- Reliable performance on voice recordings.

Disadvantages:
- Requires internet connectivity.
- API usage costs may increase with scale.
- Depends on third-party service availability.

Alternative Considered: OpenAI Whisper

Google Text-to-Speech (TTS):

Reason for Selection:
Google Text-to-Speech was selected because it offers reliable voice synthesis with simple integration and low computational requirements. It enables reconstructed memories to be delivered as audio output, improving accessibility, and usability for users.

Advantages:
- Simple and reliable implementation.
- Natural-sounding speech output.
- Supports multiple languages and voices.
- Lightweight, Open-Source, and efficient.
- Well-documented API support.

Disadvantages:
- Limited emotional expressiveness compared to premium voice services.
- Requires internet connectivity.
- Voice customization options are limited.

Alternative Considered: ElevenLabs TTS

Real-ESRGAN, GFPGAN, and BLIP (Image Processing):

Reason for Selection:
These models were selected to improve the quality and usefulness of historical photographs used during memory reconstruction. Real-ESRGAN enhances image resolution, GFPGAN restores facial details, and BLIP generates image descriptions that contribute contextual information for memory generation.

Advantages:
- High-quality image enhancement and restoration.
- Effective recovery of degraded or low-resolution photographs.
- Automatic caption generation for contextual understanding.
- Improves quality of multimedia inputs.
- Open-source and research-supported technologies.

Disadvantages:
- Computationally intensive processing.
- Processing time increases for large images.
- Caption generation accuracy depends on image quality.

Alternative Considered: OpenCV-based image enhancement techniques

Database Selection:

Selected Technology: PostgreSQL with Alembic Migrations

Reason for Selection:
PostgreSQL was selected because it provides a reliable, scalable, and feature-rich relational database management system capable of handling the structured data requirements of ReMIND. The platform stores user accounts, authentication information, memory artifacts, reminders, timelines, feedback records, audit logs, and system metadata. Alembic was selected to manage database schema migrations and version control, ensuring consistent database updates throughout development and deployment.

Advantages:
- Open-source and highly reliable.
- Strong support for relational data modeling is important.
- Excellent data integrity and consistency.
- Advanced querying and indexing capabilities.
- Scalable for future system growth.
- Alembic enables controlled and versioned database migrations.

Disadvantages:
- Requires proper schema design and maintenance.
- More complex than lightweight database solutions.
- Migration management requires careful version control.

Alternative Considered: MongoDB

Media Storage Selection:

Selected Technology: Cloudinary

Reason for Selection:
Cloudinary was selected as the primary media storage platform because it provides secure cloud-based storage, efficient media delivery, and seamless integration with web applications. Since ReMIND processes and stores user-uploaded images for memory reconstruction, Cloudinary offered a reliable solution for media management while reducing the complexity of maintaining dedicated file storage infrastructure.

Advantages
- Secure cloud-based image storage.
- Fast media delivery through CDN support.
- Easy integration with backend services and APIs.
- Automatic image optimization and transformation features.
- Reduces server storage requirements.
- Scalable storage and media management capabilities.

Disadvantages
- Storage and bandwidth limit free plans.
- Dependence on third-party service availability.
- Additional costs may arise as storage requirements increase.

Alternative Considered: Firebase Storage

Deployment Infrastructure Selection:

Selected Technology: Hostinger VPS

Reason for Selection:
Hostinger VPS was selected because it provides a dedicated and cost-effective hosting environment with full server control, allowing the complete ReMIND system to be deployed on a single

infrastructure. The platform supports backend services, database connectivity, API integrations, and web hosting requirements while offering sufficient performance for a proof-of-concept AI-assisted application. The VPS environment also provides flexibility for configuration, deployment, monitoring, and future scalability.

Advantages:
- Full administrative control over server resources.
- Supports deployment of complete application stacks on a single platform.
- Cost-effective compared to many enterprise cloud solutions.
- Dedicated computing resources improve reliability and performance.
- Flexible environment for custom configurations and software installation.

Disadvantages:
- Requires manual server configuration and maintenance.
- Security updates and system monitoring are the responsibility of administrators.
- Limited resources compared to large-scale cloud infrastructure platforms.
- Downtime recovery and backup management require additional configuration.

Alternative Considered: Render

3. Detailed/Component Design

Deployment Diagram:

The Deployment Diagram illustrates the physical architecture of the ReMIND system and the interaction between its major components. Users access the platform through a web browser, where the React.js frontend provides interfaces for memory submission, retrieval, and system interaction. All user requests are securely transmitted via HTTPS to the backend server, which is responsible for authentication, application logic, data processing, and coordination of system services. The backend integrates with several AI-powered services, including GPT-4 for memory narrative generation, Whisper for speech transcription, ElevenLabs for voice synthesis, and Real-ESRGAN for image enhancement. User accounts, memory records, and other structured information are stored in a PostgreSQL database, while multimedia content such as images, audio files, and generated artifacts are maintained in cloud storage services. After processing and reconstruction are completed, the backend delivers the generated memories, enhanced media, and relevant notifications back to the frontend for presentation to the user. This deployment architecture enables the ReMIND system to operate as a scalable, secure, and cloud-based platform powered by advanced artificial intelligence technologies.

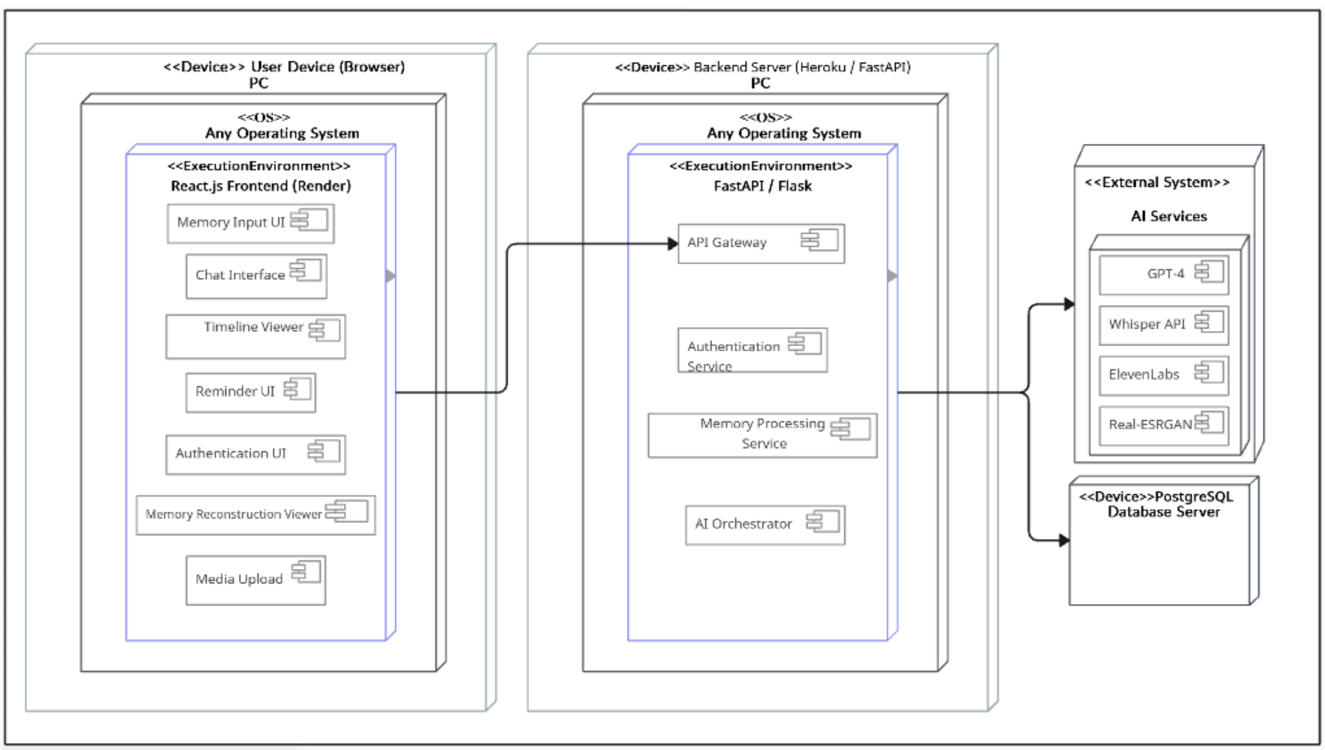


Fig 3.1: Deployment Diagram

Petri Net Diagram:

The Petri Net diagram represents the workflow of memory reconstruction within the ReMIND system, emphasizing the concurrent and synchronized execution of multiple processing activities. The process begins when users submit multimedia content, including text, images, and audio files. The input data is first validated and stored before being distributed to parallel processing stages. Separate transitions simultaneously perform text extraction and analysis, image enhancement, and audio transcription using the system's integrated AI services. The outputs generated from these independent processes are then synchronized and combined to create a comprehensive reconstructed memory. Once the reconstruction process is completed, the resulting memory artifact is stored in the system database and made available for user retrieval and viewing. The Petri Net model effectively demonstrates the system's ability to manage parallel operations, resource coordination, and process synchronization, ensuring efficient and accurate memory reconstruction.

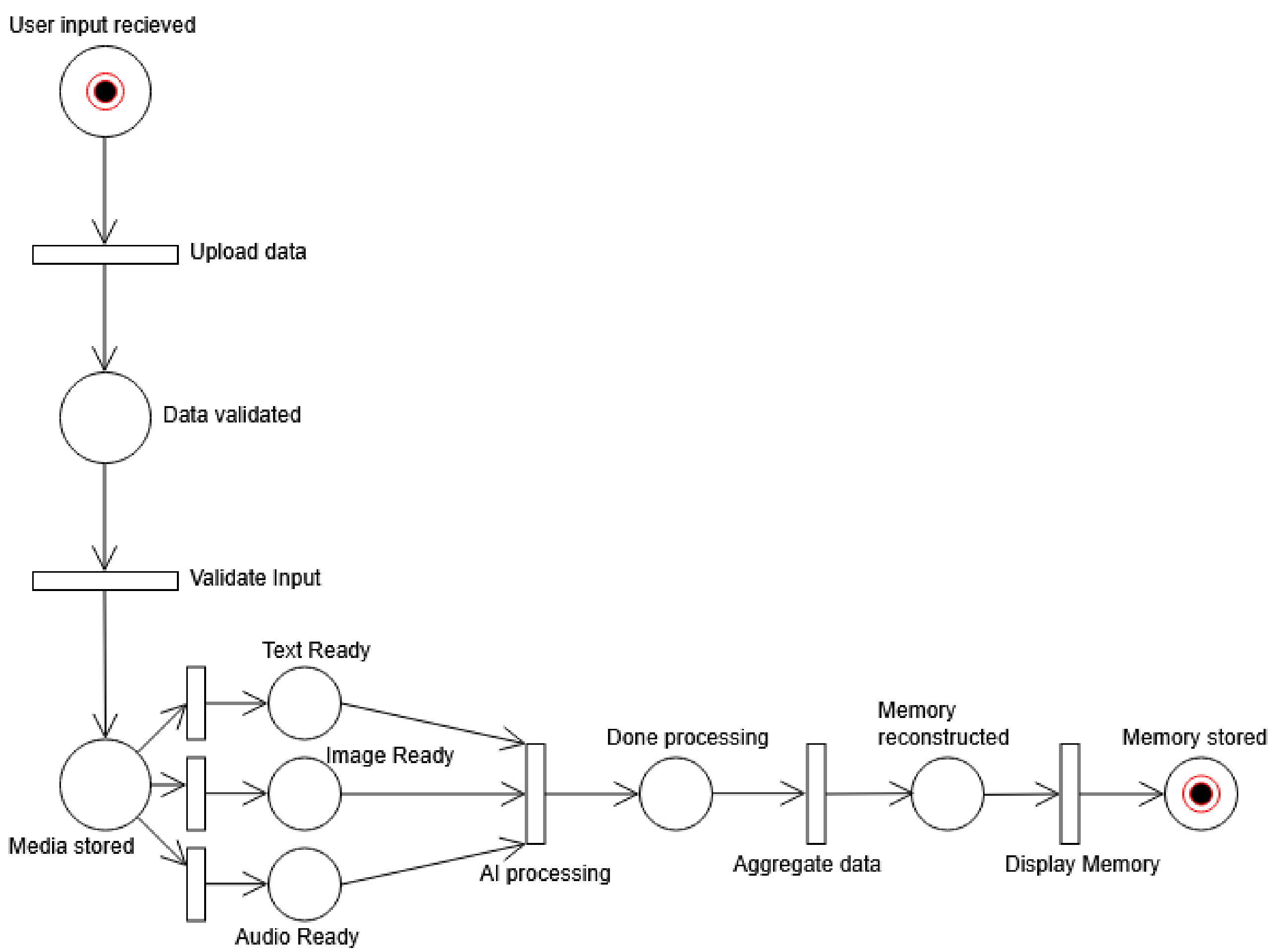


Fig 3.2: Petri Net Diagram

3.1 Component-Component Interface

The ReMIND system follows a modular architecture where internal software components collaborate to perform memory reconstruction and management tasks. The Frontend communicates with the Backend API through secure RESTful interfaces for user authentication, memory uploads, timeline retrieval, and reminder management. The Backend coordinates with the AI Orchestration Module to process text, image, and audio inputs and aggregate the generated outputs into reconstructed memories. The AI Orchestration Module interacts with specialized processing services, including text generation, speech processing, and image enhancement modules, while the Data Layer manages persistent storage of user profiles, memory artifacts, reminders, and metadata. These components exchange data through well-defined APIs and service interfaces, ensuring loose coupling, maintainability, scalability, and efficient system operation. For the corresponding diagram, refer to Figure 2.1.1 in Section 2.1.

3.2 Component-External Entities Interface

The Component Diagram illustrates the interaction between the major software components of the ReMIND system and the external services it depends upon. The Backend API serves as the central processing unit, coordinating communication between the frontend application, databases, cloud storage services, and AI-powered modules. To support memory reconstruction, the backend integrates with multiple external AI services, including GPT-4 for generating memory narratives, Whisper for audio transcription, ElevenLabs for voice synthesis, and Real-ESRGAN for image enhancement. Structured information such as user accounts, memory records, and metadata is maintained within a PostgreSQL database, while multimedia assets are stored and managed through cloud storage platforms. Additionally, notification and email services can be integrated to deliver reminders, alerts, and user updates. All interactions between system components and third-party services are conducted through secure REST APIs and HTTPS protocols, ensuring reliable, scalable, and protected communication throughout the ReMIND ecosystem.
For the corresponding diagram, refer to Figure 2.1.1 in Section 2.1.

3.3 Component-Human Interface

The Human Interface Design of ReMIND focuses on providing an intuitive, accessible, and user-friendly experience for patients, caregivers, and administrators. The system includes several key interfaces, such as the Landing Page, Login and Registration Screen, Security Questions Screen, User Dashboard, Memory Input Screen, Chat Interface, Timeline Viewer, Reconstructed Memory Viewer, Admin Dashboard, and Notification and Reminder Screen. Through these interfaces, users can securely access the platform, upload text, audio, and image-based memories, communicate with the AI assistant, and view reconstructed memories in an organized and meaningful format. The interface design follows established Human-Computer Interaction (HCI) principles to ensure usability for elderly users and individuals with limited technical expertise. Features such as large and readable typography, high-contrast color schemes, clearly labeled buttons, responsive layouts, simplified navigation, and consistent feedback messages enhance accessibility and ease of use. Special attention has been given to reducing cognitive load and creating a comfortable user experience, enabling patients and caregivers to interact with the system efficiently and confidently.

4. Screenshots/Prototype

4.1 Workflow

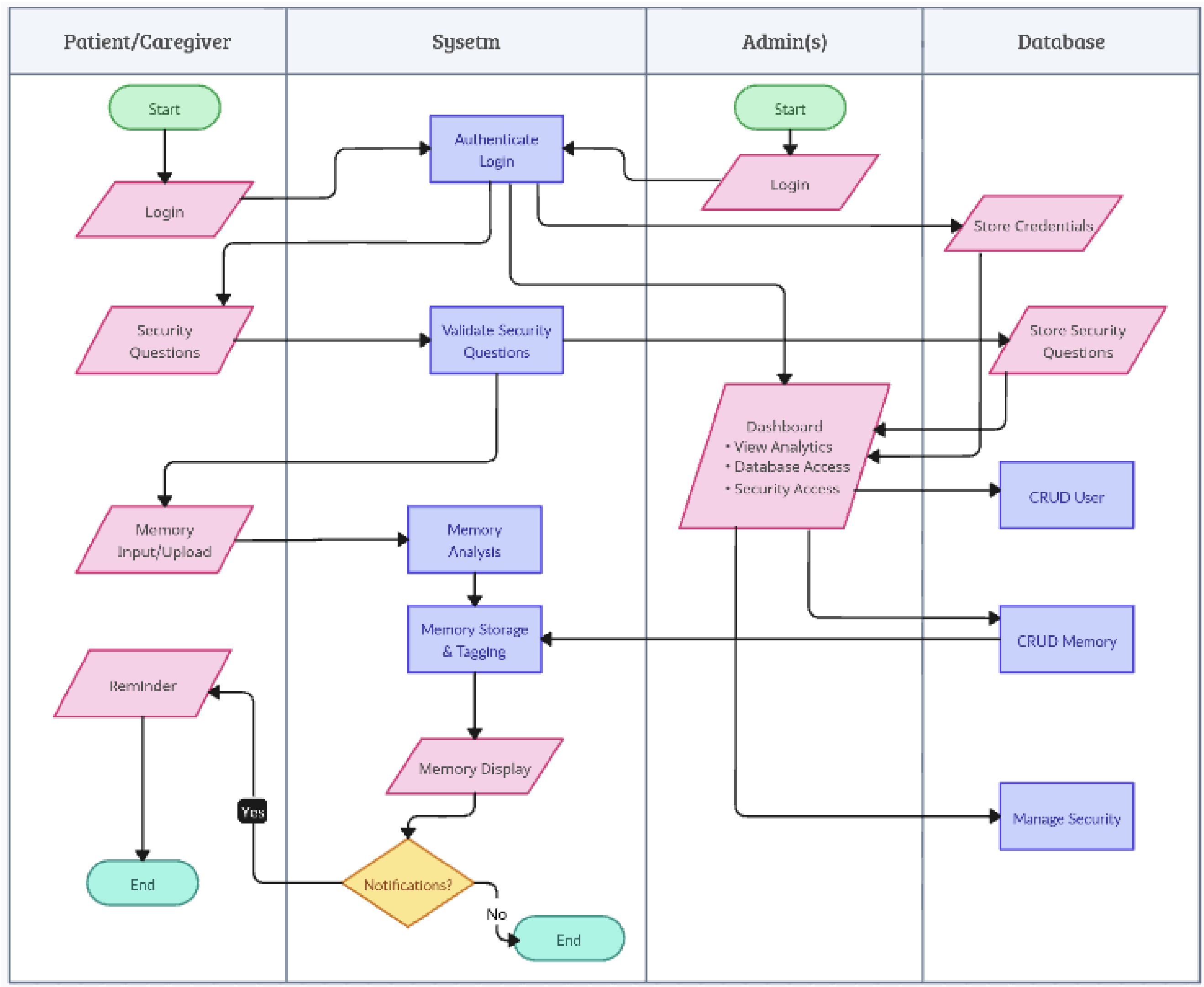
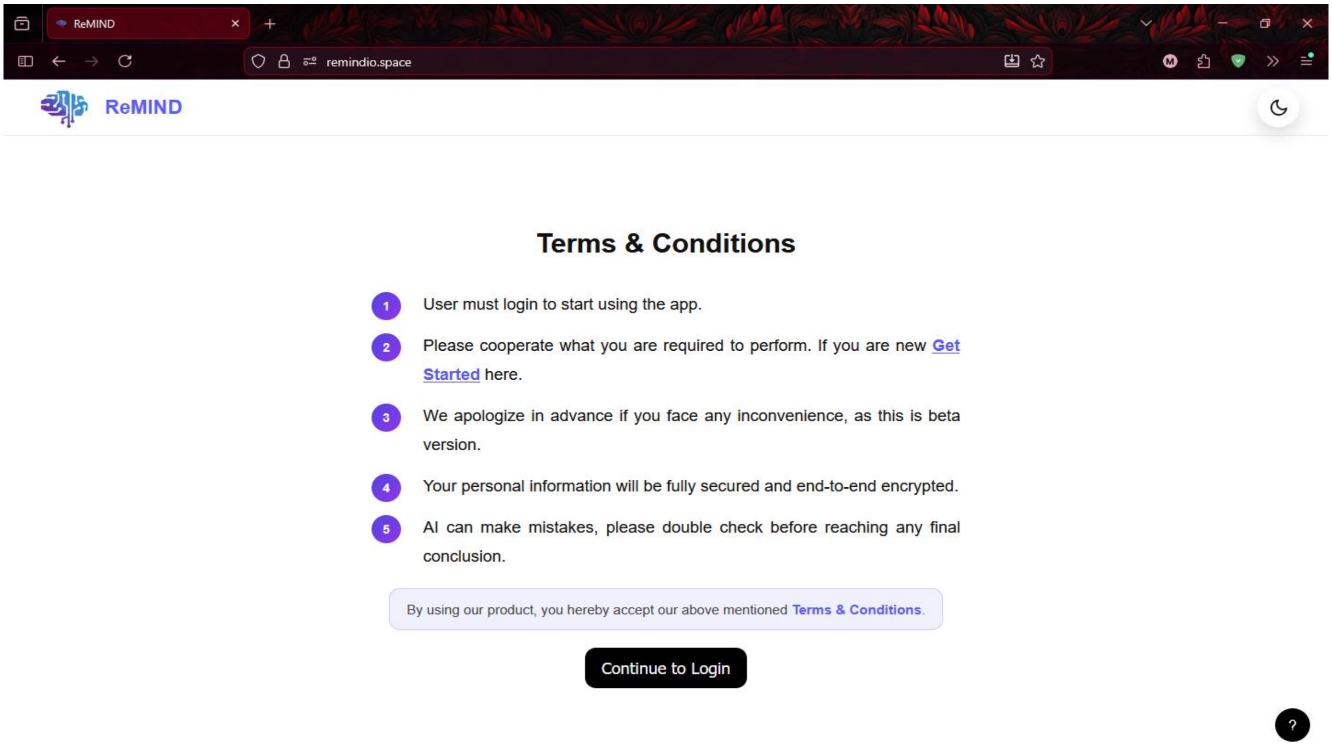
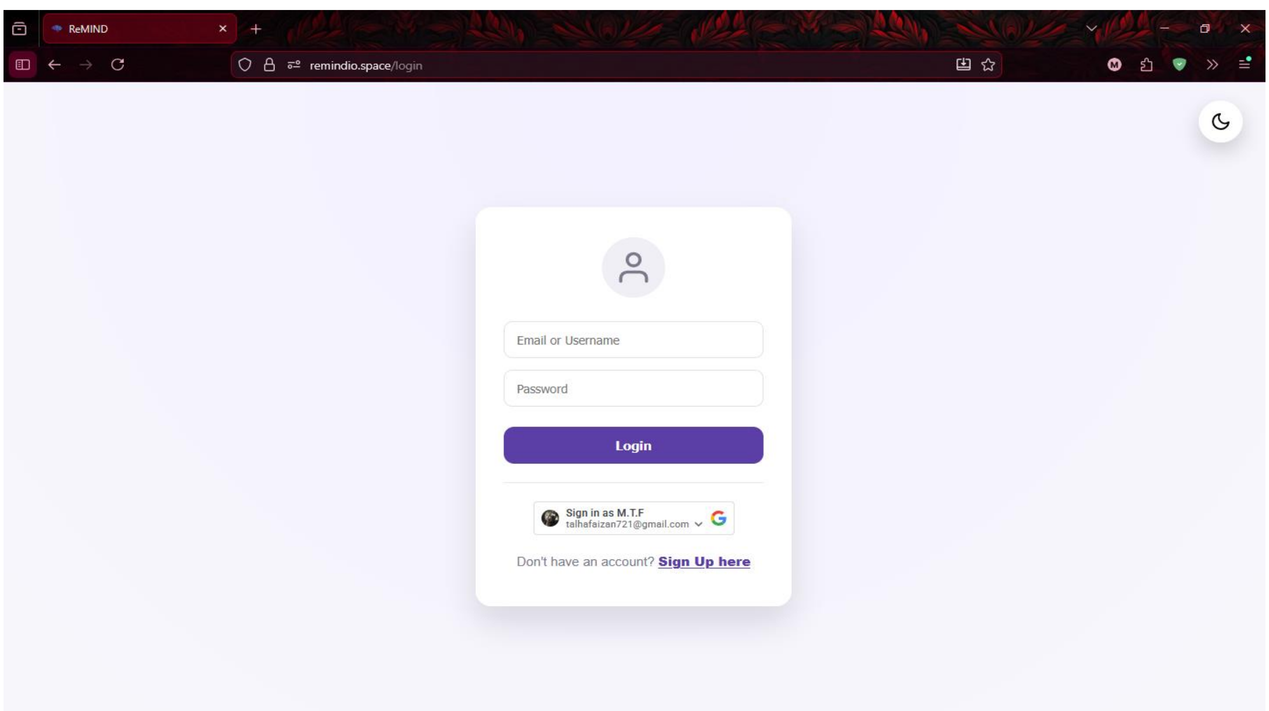
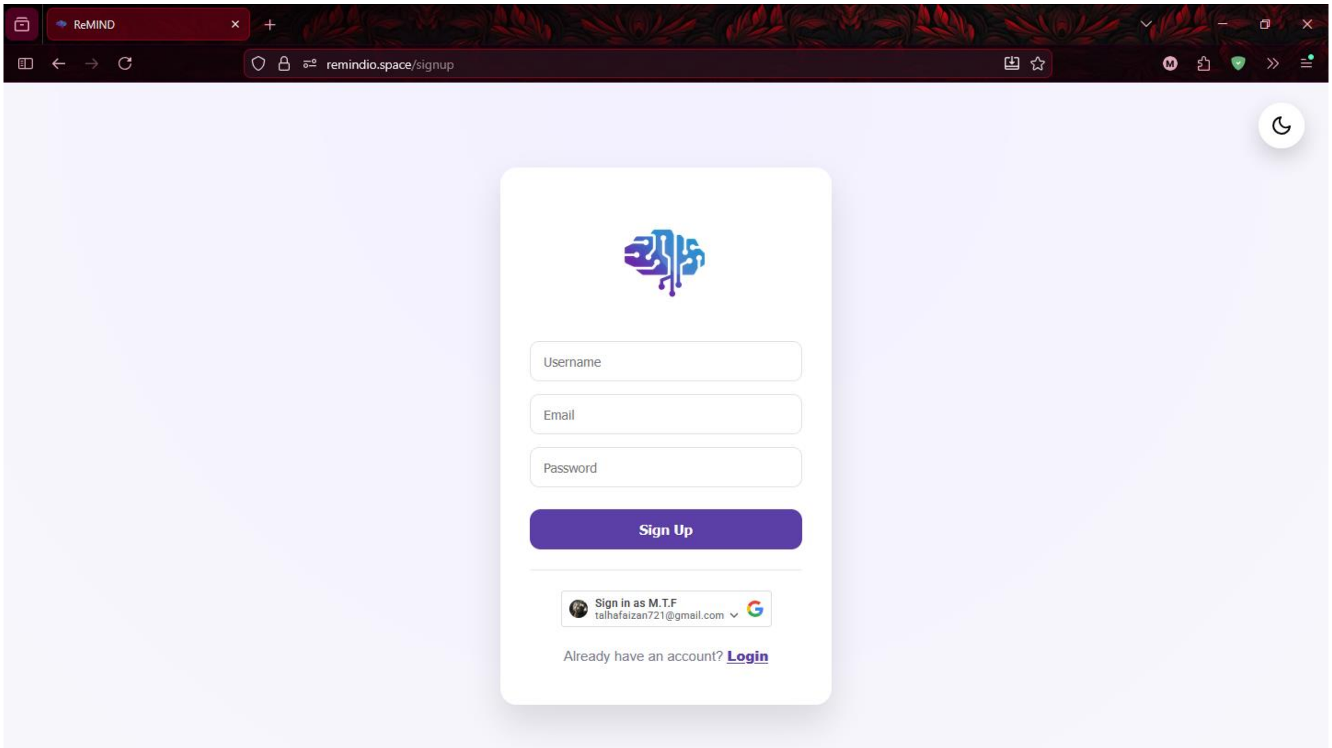


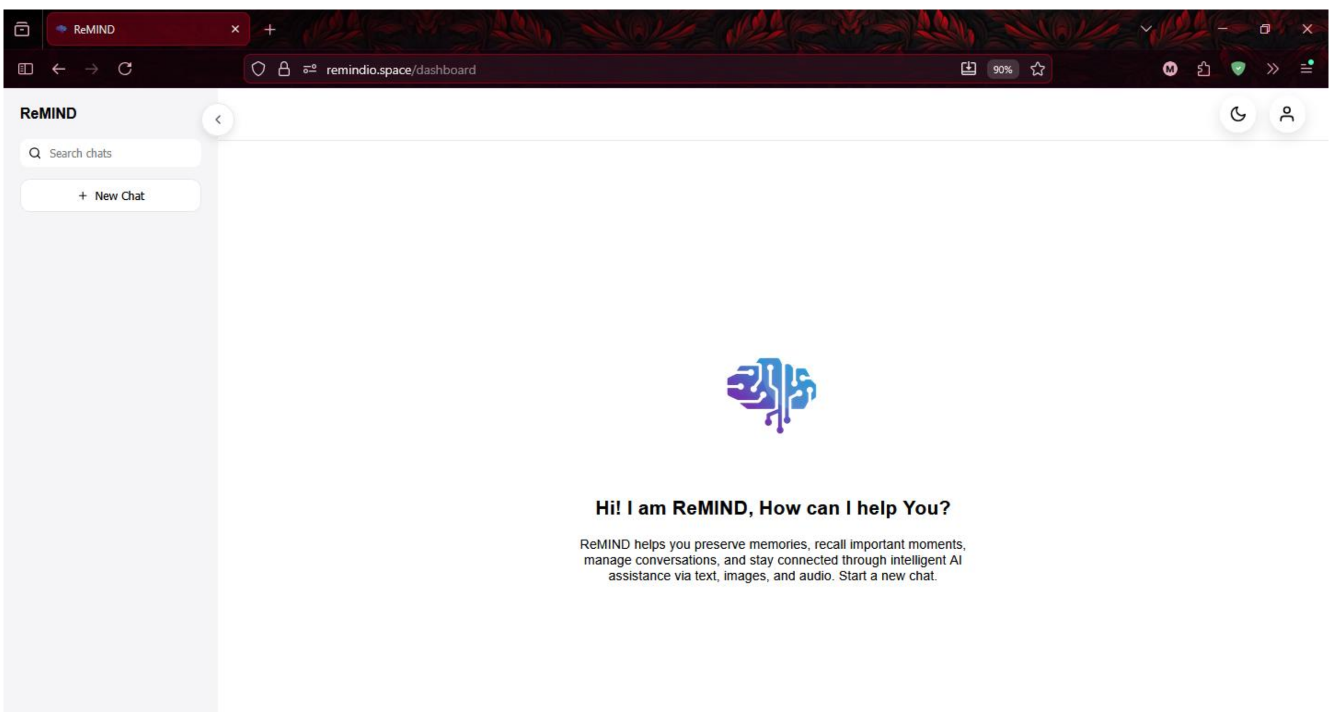
Fig 4.1: Swimlane Diagram

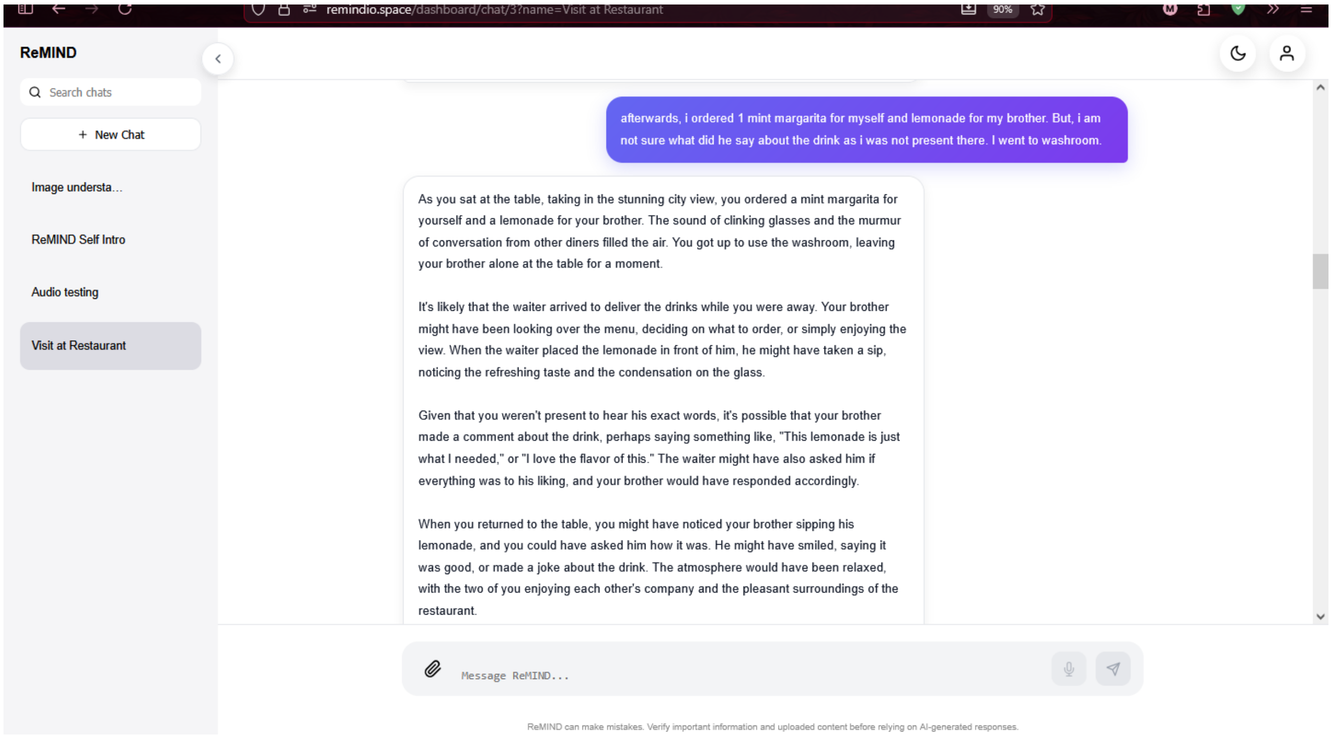
4.2 Screens


ReMIND
remindio.space
ReMIND
Terms & Conditions
1 User must login to start using the app.
2 Please cooperate what you are required to perform. If you are new Get Started here.
3 We apologize in advance if you face any inconvenience, as this is beta version.
4 Your personal information will be fully secured and end-to-end encrypted.
5 AI can make mistakes, please double check before reaching any final conclusion.
By using our product, you hereby accept our above mentioned Terms & Conditions.
Continue to Login
?


ReMIND
remindio.space/login
Email or Username
Password
Login
Sign in as M.T.F
talhafaizan721@gmail.com
Don't have an account? Sign Up here


ReMIND
remindio.space/signup
Username
Email
Password
Sign Up
Sign in as M.T.F
talhafaizan721@gmail.com
Already have an account? Login


ReMIND
remindio.space/dashboard
90%
ReMIND
Search chats
+ New Chat
Hi! I am ReMIND, How can I help You?
ReMIND helps you preserve memories, recall important moments, manage conversations, and stay connected through intelligent AI assistance via text, images, and audio. Start a new chat.



4.3 Additional Information

The ReMIND system has been developed as a proof-of-concept AI-assisted memory reconstruction platform that integrates text, image, and voice processing technologies into a unified solution. The final implementation includes secure user authentication, multimedia memory input, AI-powered memory reconstruction, timeline management, reminder functionality, and administrative controls. The system was designed with accessibility and usability in mind, particularly for elderly users and caregivers. While the current version demonstrates the feasibility of AI-driven memory reconstruction, future enhancements may include advanced emotion-aware reasoning, multilingual support, improved memory verification mechanisms, mobile application support, and integration with healthcare information systems. The project establishes a strong foundation for further research and real-world deployment in cognitive support and memory rehabilitation domains.

5. Test Specification and Results

5.1 Test Case Specification

Identifier	TC-1
Related requirements(s)	User Authentication Module
Short description	Verify that a registered user can log into the system successfully.
Pre-condition(s)	User account exists.
Input data	Valid username and password
Detailed steps	Open login page → Enter credentials → Click Login
Expected result(s)	User authenticated and redirected to the dashboard.
Post-condition(s)	JWT session created.
Actual result(s)	User successfully authenticated and dashboard loaded.
Test Case Result	PASS

Table 5.1.1: User Authentication (Login)

Identifier	TC-2
Related requirements(s)	User Authentication Module
Short description	Verify successful user registration.
Pre-condition(s)	Username and Email do not already exist.
Input data	New username, email, password
Detailed steps	Open registration page → Enter details → Submit
Expected result(s)	User account created successfully.
Post-condition(s)	User records stored in PostgreSQL.
Actual result(s)	The user account was created successfully.
Test Case Result	PASS

Table 5.1.2: User Registration

Identifier	TC-3
Related requirements(s)	Memory Reconstruction Module
Short description	Verify memory generation from text input.
Pre-condition(s)	The User logged in.
Input data	Textual memory fragment
Detailed steps	Enter text → Submit
Expected result(s)	AI-generated memory narrative returned.
Post-condition(s)	Memory is stored in the database.
Actual result(s)	The memory was reconstructed successfully.
Test Case Result	PASS

Table 5.1.3: Text Memory Reconstruction

Identifier	TC-4
Related requirements(s)	Multimedia Input Module
Short description	Verify image upload and processing workflow.
Pre-condition(s)	The User Logged in.
Input data	Valid Format Image
Detailed steps	Upload image → Submit
Expected result(s)	Image processed and metadata generated.
Post-condition(s)	Image stored in Cloudinary.
Actual result(s)	Image processed successfully.
Test Case Result	PASS

Table 5.1.4: Image Upload and Processing

Identifier	TC-5
Related requirements(s)	Voice Processing Module
Short description	Verify audio upload and speech processing.
Pre-condition(s)	The User logged in.
Input data	Valid Format Audio File
Detailed steps	Upload Audio -> submit
Expected result(s)	Speech converted to text and processed.
Post-condition(s)	Voice data was processed successfully.
Actual result(s)	Audio transcription was generated successfully.
Test Case Result	PASS

Table 5.1.5: Voice Upload and Processing

Identifier	TC-6
Related requirements(s)	Memory Management Module
Short description	Verify retrieval of previously generated memories.
Pre-condition(s)	Existing memories are available.
Input data	User requests.
Detailed steps	Open memory history page.
Expected result(s)	Previously generated memories were displayed.
Post-condition(s)	Memory records were retrieved successfully.
Actual result(s)	Memory history loaded correctly.
Test Case Result	PASS

Table 5.1.6: Memory History Retrieval

Identifier	TC-7
Related requirements(s)	Application Architecture
Short description	Verify successful API communication.
Pre-condition(s)	Backend running.
Input data	API request.
Detailed steps	Send request → Receive response
Expected result(s)	HTTP 200 responses were returned.
Post-condition(s)	Data exchanged successfully.
Actual result(s)	Successful communication is established.
Test Case Result	PASS

Table 5.1.7: Frontend Backend Communication

Identifier	TC-8
Related requirements(s)	Database Management Module
Short description	Verify data persistence and retrieval.
Pre-condition(s)	Database is available.
Input data	User and memory records.
Detailed steps	Insert → Retrieve → Verify
Expected result(s)	Data stored and retrieved correctly.
Post-condition(s)	Database consistency maintained.
Actual result(s)	Database operations are executed successfully.
Test Case Result	PASS

Table 5.1.8: Database Operations

Identifier	TC-9
Related requirements(s)	Media Storage Module
Short description	Verify image storage in Cloudinary.
Pre-condition(s)	Cloudinary configured.
Input data	Image file
Detailed steps	Upload image
Expected result(s)	File stored and URL generated.
Post-condition(s)	Media accessible through Cloudinary.
Actual result(s)	Image stored successfully.
Test Case Result	PASS

Table 5.1.9: Cloudinary Media Storage

Identifier	TC-10
Related requirements(s)	End-to-End System Integration
Short description	Verify complete system workflow from input to memory generation.
Pre-condition(s)	The user logged in.
Input data	Text, image, and voice inputs.
Detailed steps	Upload inputs → Process → Generate memory
Expected result(s)	Unified reconstructed memory generated.
Post-condition(s)	Memory stored and displayed.
Actual result(s)	The end-to-end workflow was completed successfully.
Test Case Result	PASS

Table 5.1.10: Complete Memory Reconstruction Workflow

5.2 Summary of Test Results

Module Name	Test cases run	Number of defects found	Number of defects corrected so far	Number of defects still need to be corrected
Authentication	TC1, TC2	3	3	0
Memory Reconstruction	TC3, TC10	6	6	0
Image Processing	TC4	4	4	0
Audio Processing	TC5	5	5	0

Memory Management	TC6	1	1	0
API Communication	TC7	8	8	0
Database Management	TC8	6	6	0
Media Storage	TC9	1	1	0
Complete System	TC1 - TC10	34	34	0

Table 5.2: Test Case Summary

6. Project Completion Status

Module Name	Status
User Authentication Module (Login / Signup / Session Management)	Complete
User Interface Module (React Frontend UI, Navigation, Dashboard)	Complete
Memory Input Module (Text, Image, and Audio)	Complete
Multimedia Processing Module (Image Enhancement, Face Restoration, Image Captioning)	Complete
Speech Processing Module (Speech-to-Text, Audio Processing, Voice Synthesis)	Complete
Memory Reconstruction Module (Context-Aware AI Narrative Generation)	Complete
Backend API Module (FastAPI Services and System Integration)	Complete
Database Management Module (PostgreSQL, SQLAlchemy, Alembic Migrations)	Complete
Cloud Media Storage Module (Cloudinary Integration)	Complete
Administration and Security Module (RBAC, Audit Logs, Protected Routes)	Complete
Testing and Validation Module	Complete
Event Clustering and Timeline Management Module	Partially Implemented
Reminder and Notification Module	Not Complete
Complete System	**Complete**

Table 6.1: Project Completion Status

Target/Objective	Status	Reason(s)
Memory Reconstruction	Completed	
Reminder and Notifications	Not Completed	Due to project scope and time constraints, reminder scheduling and notification functionality were not implemented in the final deployed system.
Elderly Friendly Interface	Completed	
Secure Data Handling	Completed	
Proof of Concept Demonstration	Completed	
Caregiver Collaboration	Partially Completed	Anyone on behalf of a patient from user account can access and use the system, but specific separate caregiver role-based integration was not fully implemented due to time constraints.
Number of Targets Completed	4/6	
Number of Targets Partially Completed	1/6	
Number of Targets Not Completed	1/6	

Table 6.1: Project Completion Status

7. Deployment/Installation Guide

Accessing the Deployed System:

ReMIND is currently deployed and publicly accessible through its production URL: **https://remindio.space**

Users can access the platform directly through a modern web browser without requiring any local installation. The deployed version of the system is scheduled to remain available until **1 July 2026**.

System Requirements:

- Modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari)
- Stable internet connection
- Microphone access for voice-based features
- Device capable of uploading image and audio files

Local Installation (After Deployment Period):
After the public deployment period ends, the project can be accessed and installed locally through the official GitHub repository.

Prerequisites:
- Python 3.10 or later
- Node.js and npm
- PostgreSQL and Git
- Internet connection
- Required API credentials

Backend Setup:
- Clone the project repository.
- Navigate to the backend directory.
- Create and activate a Python virtual environment.
- Install project dependencies.
- Configure environment variables and API credentials.
- Apply database migrations using Alembic.
- Start the FastAPI server.

Frontend Setup:
- Navigate to the frontend directory.
- Install required npm packages.
- Configure environment variables if required.
- Start the React application.

Database Setup:
1. Install PostgreSQL.
2. Create the required database.
3. Configure database connection settings.
4. Execute database migrations.

Verification:
After successful installation:
- The frontend interface should be loaded correctly.
- User registration and authentication should function properly.
- Text, image, and voice reconstruction features should operate successfully.
- Database connectivity should be established.
- AI services should generate memory reconstruction outputs correctly.

8. User Manual

Introduction:

ReMIND is an AI-assisted memory reconstruction platform that helps users reconstruct meaningful memories from fragmented text, image, and voice inputs. The system combines multiple AI technologies to generate context-aware memory narratives through a unified web interface.

Registration and Login:

1. Open the ReMIND web application.
2. Register a new account or sign in using existing credentials.
3. Alternatively, use Google Authentication if available.
4. Upon successful authentication, the system redirects the user to the main dashboard.

Text-Based Memory Reconstruction:

1. Navigate to the memory reconstruction interface.
2. Enter a memory fragment, note, description, or contextual information.
3. Submit the input for processing.
4. Review the generated memory narrative returned by the system.

Image-Based Memory Reconstruction:

1. Select the image upload option.
2. Upload an image from your device.
3. Wait for image enhancement and analysis.
4. Review the generated visual description and reconstructed memory output.

Voice-Based Memory Reconstruction:

1. Select the voice input option.
2. Record audio or upload a supported audio file.
3. Submit the recording for processing.
4. Review the generated transcription and reconstructed memory narrative.

Viewing Generated Memories:

1. Access to the memory history section.
2. Browse previously generated memory reconstructions.
3. Select a memory to view its complete details and associated outputs.

Administration Features:

Authorized administrators can:

1. View registered users.
2. Monitor system activity.
3. Access platform analytics.
4. Review audit logs.
5. Manage system records.

Logout:

1. Open the navigation menu.
2. Select the Logout option.
3. The system will terminate the active session and redirect the user to the login page.

Troubleshooting:

Unable to Upload Media:
- Verify internet connectivity.
- Ensure supported file formats are being used.
- Confirm that uploaded files are not corrupted.

Voice Features Not Working:
- Verify microphone permissions.
- Check the browser audio settings.
- Ensure a stable internet connection.

Account Access Issues:
- Verify the entered username/email and password.
- If account information needs to be updated or access issues persist, contact the ReMIND administrators.
- Users may also refer to the project's GitHub repository or contact the development team through the official project email for assistance.

System Unavailable:

The public deployment is scheduled to remain available until **1 July 2026**. After this period, users can access the source code through the GitHub repository and deploy the system locally following the installation guide provided above.

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10. Project Summary Form

Name of Project	ReMIND: Generative AI based Memory Reconstruction
Project Type	Final Year Project (FYP)/AI based Web Application
Department	Faculty of Information Technology & Computer Science
Start Date	May 2025
Completion Date	June 2026
Supervisor / Team Leader	Supervisor: Muhammad Zulkifl Hassan Team Lead: Muhammad Hassan Ashraf
Team Members (if any)	Muhammad Hassan Ashraf, Muhammad Talha Faizan, Mubashar Tanveer
Funding Agency (if any)	Self-Funded
Amount of Funding (if any)	Approx. PKR 25000
Assign SDGs to Project	SDG 3: Good Health and Well-Being SDG 9: Industry, Innovation and Infrastructure
Motivation of Project	The idea behind ReMIND originated from a personal experience. One of our team members witnessed the challenges faced by his father due to Alzheimer's disease and the emotional impact it had on both the patient and the family. This experience helped us understand the importance of preserving memories and supporting individuals affected by memory-related disorders. Motivated by this challenge, we decided to utilize the power of Artificial Intelligence for human welfare and mental healthcare. Our goal was to develop a solution that not only demonstrates the capabilities of modern AI but also contributes positively to society by helping patients reconnect with their memories and improving their overall quality of life.
Practical / Potential Application	Cognitive assistance for individuals experiencing memory loss, memory preservation systems, assisted living environments, elderly care support, healthcare technology solutions, and future digital therapeutic platforms.
Abstract	ReMIND is an AI-assisted memory reconstruction platform designed to help users rebuild meaningful memories from fragmented information. The system integrates text processing, voice processing, and image enhancement technologies within a unified web application. Using multimodal artificial intelligence techniques, ReMIND analyzes user-provided text, images, and audio recordings to generate context-aware memory narratives. The platform incorporates secure authentication, cloud-based storage, conversational interaction, and memory reconstruction workflows to provide an accessible and user-friendly experience. Developed as proof-of-concept, ReMIND demonstrates the potential of intelligent memory assistance systems in supporting cognitive well-being and future healthcare innovation.
Key Technical Features	• Multimodal Memory Reconstruction (Text, Voice, Image) • Retrieval-Augmented Generation (RAG) • Groq Llama 3 Integration • Speech-to-Text and Voice Synthesis • Image Enhancement and Face Restoration • AI-Based Image Captioning • JWT Authentication and Google OAuth • Role-Based Access Control

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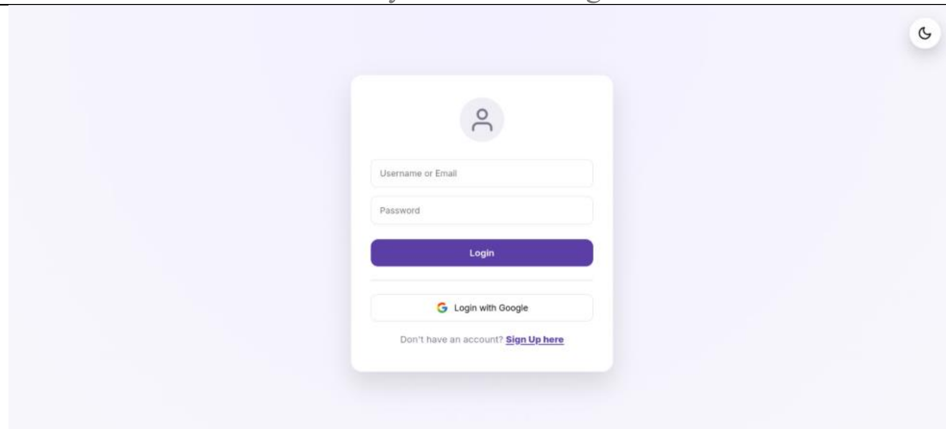
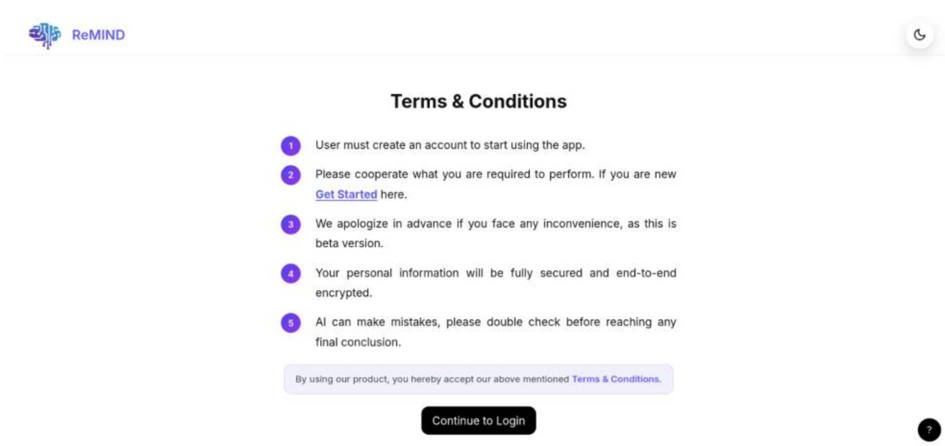
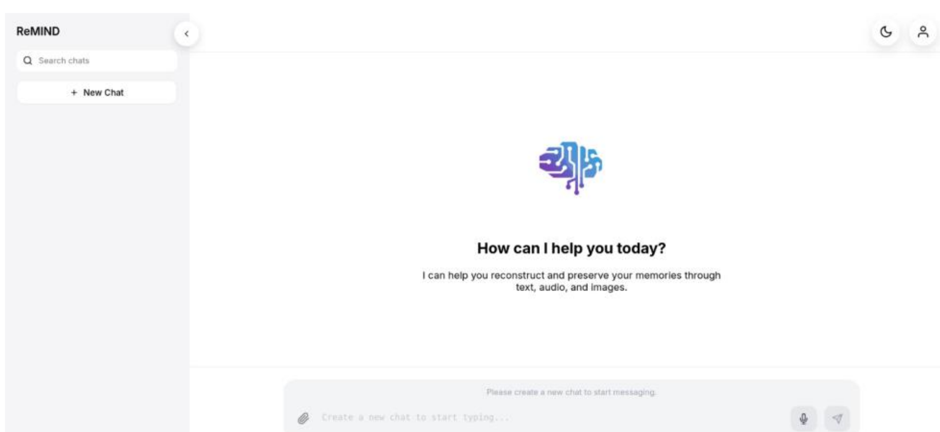
• PostgreSQL Database Management
• Cloudinary Media Storage
• FastAPI Backend Architecture
• React.js Responsive Frontend
• Admin Dashboard and Analytics Monitoring
</td>
</tr>
<tr>
<td>Projects Images / Screenshots</td>
<td>







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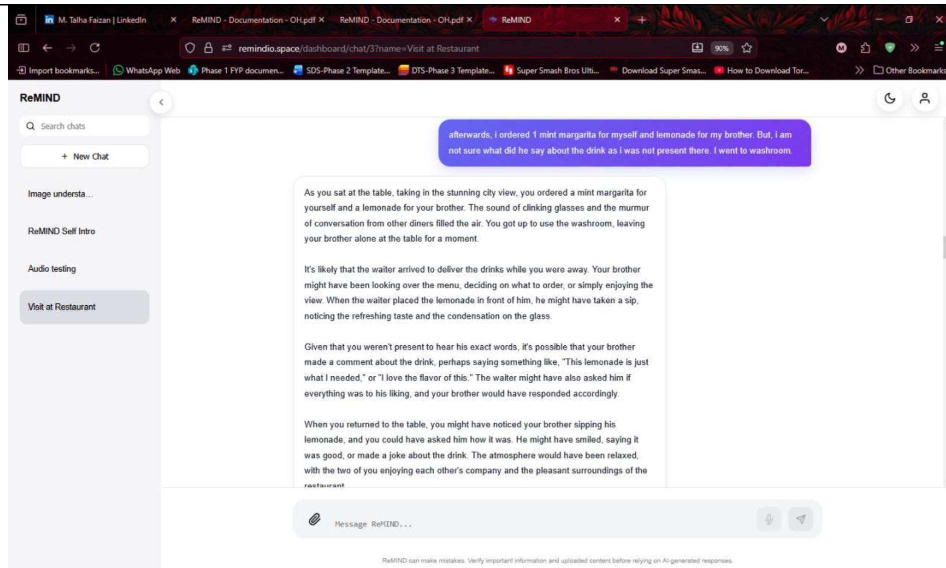
ReMIND
Search chats
+ New Chat
Image understa...
ReMIND Self Intro
Audio testing
Visit at Restaurant
afterwards, i ordered 1 mint margarita for myself and lemonade for my brother. But, i am not sure what did he say about the drink as i was not present there. I went to washroom.
As you sat at the table, taking in the stunning city view, you ordered a mint margarita for yourself and a lemonade for your brother. The sound of clinking glasses and the murmur of conversation from other diners filled the air. You got up to use the washroom, leaving your brother alone at the table for a moment.
It's likely that the waiter arrived to deliver the drinks while you were away. Your brother might have been looking over the menu, deciding on what to order, or simply enjoying the view. When the waiter placed the lemonade in front of him, he might have taken a sip, noticing the refreshing taste and the condensation on the glass.
Given that you weren't present to hear his exact words, it's possible that your brother made a comment about the drink, perhaps saying something like, "This lemonade is just what I needed," or "I love the flavor of this." The waiter might have also asked him if everything was to his liking, and your brother would have responded accordingly.
When you returned to the table, you might have noticed your brother sipping his lemonade, and you could have asked him how it was. He might have smiled, saying it was good, or made a joke about the drink. The atmosphere would have been relaxed, with the two of you enjoying each other's company and the pleasant surroundings of the
Message ReMIND...

Appendix A: Glossary

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table B.1: List of non-trivial defects

This document has been adapted from the following:

1. Previous project templates at UCP
2. High-level Technical Design, Centers for Medicare & Medicaid Services. (www.cms.gov)